The first sensitivity analysis read against the literature (discussion for 20 November)

Written on 8 November 2026 (unattended session, branch `feature/analysis` cut from `feature/sensitivity` at `55a8d9e`), for the group to read before the simulated seminar of 20 November and the choice of the fixed version (step 5 of `review-v2.md`). It reads `sensitivity-results.md` against the texts the group works from, illustrates each finding with seed-level runs, and goes further where the results leave a hole (Part B). Nothing in it is a decision: section 4 proposes rows for `choices.md`.

How to read the numbers. Every number carries its cell, population and seeds, written `[stage, population, seeds]`; `mixed` means label "shaikh" on the unit-cost rule (h = 0) and label "okishio" on the profit-rate rule (h = best rate at the firm's price), and the two neutral controls put both labels on one rule (`all_unit_cost`, `all_profit_rate`). *Share won* is the share of decided seeds in which the label "shaikh" (the unit-cost rule) holds more than half of producing capacity at t = 199, with its 95 % Wilson interval; *beyond drift* means the mixed interval is disjoint from both controls' intervals. All cell numbers come from `outputs/sensitivity/cells.csv`, seed numbers from `runs.csv` (row = `run_id`), new tables and figures from `outputs/analysis/`. Page numbers refer to the texts listed in the appendix; a claim about a text I could not open is marked *(from memory, not checked)*.

Reproducibility note (new). The rows of `runs.csv` are reproduced exactly by `experiments.sensitivity rerun` only on Python 3.12 or later: Python 3.12 changed `sum()` on floats to compensated summation, so on Python 3.10 a rerun of `1_base_mixed_s4` diverges in 36 observables (checked on 8 November; on 3.13 it reproduces the row exactly). CI runs 3.11; `requirements.txt` should pin `python_requires >= 3.12` or the README should say so (proposed row in section 4). A second, smaller trap: pandas' default CSV float parser is not exact on every platform (on x86_64 it loses the last digit of some values, on the Mac it does not), so `experiments.sensitivity.load_runs` now reads with `float_precision="round_trip"`. The stage 4 rows of this session were computed on Linux (Python 3.13.18); of five stage 1–3 rows checked on that machine, four reproduce exactly and one (`1_is_neutral_mixed_s7`) differs by 2.2e-16 in one summary observable (`ucs_spread_mean`), the trajectory being identical.

Session note. The connection to the Mac dropped at about 01:30 UTC, after the Part A commit and after the B3 candidate runs had been written on the Mac, and came back at 04:40 UTC. In between, stage 4 (B2) was computed in the session's container (Linux, x86_64) on the model code of `55a8d9e`, which the candidate switches do not change at their defaults, and the B3 cells were run a second time there. Back on the Mac's VM, four stage 4 rows (`4_anchor_r10_mixed_s3`, `4_lf-acc-g01-cat-r16_mixed_s5`, `4_is15-tau_010_mixed_s2`) and `1_base_mixed_s4` were rerun after the merge: all reproduce their rows exactly.

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Appendix: texts opened, with the pages used

1. The question and what the literature predicts

The research question (`CLAUDE.md`): in an industry where cheaper techniques often need more capital per unit of output, does market selection favour the firms that equip new plants at the lowest unit cost (Shaikh, h = 0) or at the highest profit rate at the current price (Okishio, h = best rate on offer), and how does the answer depend on price wars and on how investment is financed? The table turns the literature into claims that the results can meet or fail. Observables are those of `sensitivity-plan.md`, section 3.

#	Claim	Who, where	Observable(s)	What the results say	Verdict
1	In real competition the lowest-cost capitals are the price leaders; followers' margins are residuals; the lowest-cost condition has the highest *sustainable* profit rate under price cutting, so the lowest-unit-cost technique displaces a rival even when it earns less at the going price (D2 against C at $100)	Shaikh 2016, Table 7.3 rows 1–5 (p. 271); pp. 313–317 (Tables 7.10–7.12, eq. 7.11, fig. 7.22)	share won; `leader_lowest_cost_share`; `d_share_price_all`; `price_to_pp`	`base` [3, mixed, 1–500] 0.81 (0.77–0.84) against 0.48 / 0.50; the leader is the lowest-cost firm in 0.94 of periods; price term of the sales share +0.25; 86 of 120 factorial cells beyond drift, all for the unit-cost rule	**Met**, except in two corners (A.7); but with the version 1 scrapping test (operating cost) the base advantage vanishes (0.55, B2)
2	Under real competition a capital-using, cost-reducing technique spreads by price cutting and the rate of profit falls even at a given real wage; under the profit-rate criterion it falls only at high wage shares	Shaikh 2016, p. 261; pp. 317–321 (Alternative 2: general rate 16 % → 15.2 %); Michl 2023, pp. 47–48	`marx_biased_*`; `profit_rate_total` of the two controls	Spreads: realised change is Marx-biased for the unit-cost label in 0.84 of `base` seeds. Rate: an industry where every firm uses the unit-cost rule realises a lower profit rate than one where every firm uses the profit-rate rule (`is_marx_15` [1, controls, 1–200]: 8.5 % against 10.9 %), at similar prices (A.6)	**Met at industry level**; the economy-wide claim is **not testable** (rbar exogenous)
3	After a cost-cutter takes over, the price first falls, then rises to the new regulating price of production (D2: back up to $101.30)	Shaikh 2016, p. 316	`price_rise_after_adoption`; `price_to_pp`; figure 5	Real competition: the price at t = 199 exceeds that at t = 100 in 0.17–0.31 of seeds (innovation set, marx 0.5 to 3) and ends 10–20 % above the price of production; leader-follow: at most 0.06, ending below it (0.85–0.93)	**Met under real competition only**
4	A technique that raises the profit rate at current prices cannot lower the general rate once new prices of production settle, at a constant real wage (Okishio's theorem)	Okishio 1961 *(not in the library)*, as stated in Shaikh 2016, p. 320; Shaikh 1978, pp. 241–242; Roemer 1979, p. 379; Duménil & Lévy 2005, p. 209	`profit_rate_okishio` against `profit_rate_shaikh`; controls' `profit_rate_total`	The theorem is about the general rate: **not testable here**, since rbar is a parameter and the money wage is the numéraire (the constant real wage holds by construction, the general rate does not respond). Its micro premise, that the adopter earns the higher rate, fails: `base` 4.5 % (profit-rate label) against 8.4 %	**Not testable**; premise **failed**
5	A technique that does not pass the profit-rate test is never adopted by a rational capitalist; a higher-cost method can dominate if it has the lower price of production (Sraffa's criterion)	Shaikh 2016, p. 320 n. 36 and p. 322 (stating the standard view); Roemer 1979, p. 379	`conflict_adopted_shaikh`; `okishio_dearer`; share won	The unit-cost firms adopt every conflict-zone design (`conflict_adopted_shaikh` 1.0) and they are the ones selected; the profit-rate firms' refusals and dearer orders go with losing (`okishio_dearer` 0.49 in seeds the unit-cost rule wins, 0.17 where it loses, stage 2)	**Failed** as a statement about who survives
6	The two criteria conflict only when technical change is Marx-biased (capital-using, labour-saving) and when the rule is evaluated at prices that the adoption will change; the profit-rate criterion is the right one if prices do not fall after adoption	Duménil & Lévy 2005, p. 209; Michl 2023, pp. 47–48; Shaikh 2016, pp. 323–324	zone shares; `marx_biased_*`, `gamma_*`, `chi_*`; `no_tech`; the neutral set; anchor cells	Fixed menu: drift (0.51). Neutral set: conflict share 0.04, share won 0.78. Anchor pricing (prices held at each firm's target return) at rbar 0.16: the profit-rate rule wins (0.26)	**Met**

#	Claim	Who, where	Observable(s)	What the results say	Verdict
7	Selection on realised profitability drives accumulation, selection on price drives market share; the balance depends on finance and on the intensity of selection	Nelson & Winter 1982, pp. 17, 19; Duménil & Lévy 1995, p. 217; Metcalfe 1998 *(not in the library)*	`d_share_accum_all` against `d_share_price_all`; the finance arm; CHI	The price term is the larger in 93 % of cells; accumulation decides only with self-finance, growing demand and no entry, where the investment rule flips the winner (`lf-re-g01-cat-r16` 0.32 → 0.85 with the accelerator [3, mixed, 1–500]; no seed borrows); CHI 0.25–4 moves nothing (0.71–0.79)	**Met for the investment rule; not for selection intensity** (A.8)
8	Selection is weak in the data: relative productivity explains little of firm growth, and survival turns on profitability and demand more than on physical productivity	Dosi et al. 2015 (abstract, introduction); Foster, Haltiwanger & Syverson 2008, pp. 394, 397	CHI 0.25 cells	χ = 0.25 delays takeovers (mean `takeover_t` 174 against 117) without changing the winner (0.71 [1, mixed, 1–200])	Not a test of the data; **the briefs' χ trap does not spring**
9	Vintages coexist; old plants run until their operating cost reaches the total cost of best practice (normal return included), so the vintage spread and the replacement margin set the pace of diffusion	Salter 1966, pp. 56–57, 64–65, 84–85	`ucs_spread_mean`; `capital_age_*`; `replacements_*`	Spread: 1.23 (`base`); replacements ≈ 0 everywhere at the defaults; capital ages equal in `base` (14.0 / 13.9) but 32.6 against 22.7 for the losing unit-cost label in `lf-re-g01-cat-r16`	**Partly met**; the replacement margin is never reached
10	Entry equalises the profit rate on new investment in the regulating capitals across industries, and is how the winning practice spreads	Shaikh 2016, pp. 265, 296 (citing Botwinick 1993, ch. 5, *not in the library*)	`entries`, `d_share_entry_all`, `r_new_mean`, `periods_above_rbar`; rbar 0.16 cells	rbar 0.05: `r_new_mean` 0.054 against rbar 0.05, the industry rate above rbar in 95 of 100 periods, entrants inherit the majority rule; rbar 0.16: the rate is above rbar in 0.3–2.3 of 100 periods, entry stops, and both reversals live there	**Met at rbar 0.05**; at 0.16 the carrier is shut
11	Diffusion takes time and is shaped by learning and by who knows what	Silverberg, Dosi & Orsenigo 1988, p. 1032	`is_private` against `is_marx_15`	0.87 (0.82–0.91) against 0.94 (0.90–0.97): overlapping	**No effect** on the winner

2. Part A: the findings read against the literature

Figure A0 (`outputs/analysis/figures/A0_summary_section6.png`) places every cell of section 6 of the results against its controls: the grey band is the union of the two controls' intervals, a coloured dot is beyond drift (blue for the unit-cost rule, orange for the profit-rate rule), a grey dot is not.

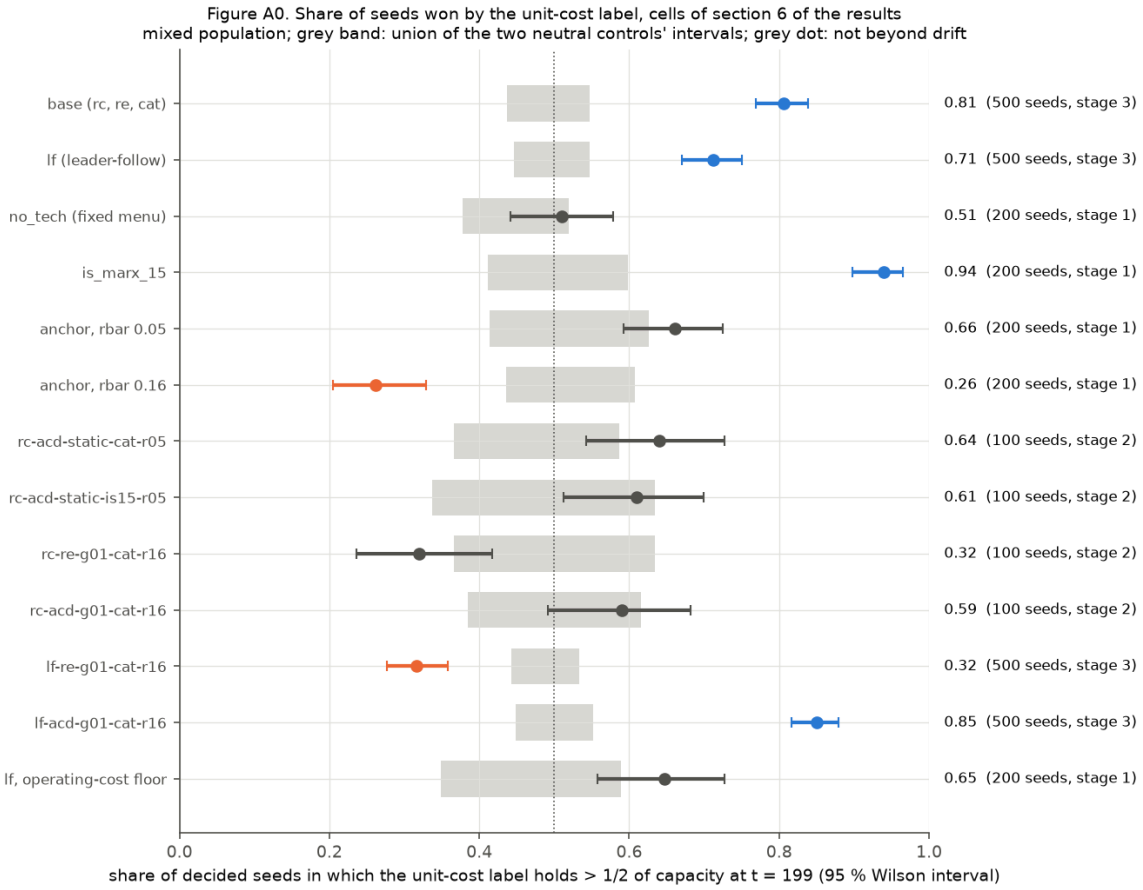


Figure A0 — Share of seeds won by the unit-cost label in the cells of section 6 of the results (mixed population); grey band: union of the two neutral controls' intervals; grey dot: not beyond drift.

A.1 The base result and its controls

Result. `base` [3, mixed, 1–500]: the unit-cost rule wins 0.81 (0.77–0.84) of seeds; controls 0.48 (all unit cost) and 0.50 (all profit rate); mean final capacity share of the unit-cost label 0.80; no undecided seed. The leader-follow arm `lf` [3, mixed, 1–500]: 0.71 (0.67–0.75), controls 0.50 / 0.49. Both beyond drift, and the 10-point gap between the two pricing rules is itself beyond the intervals.

Literature. Row 1: Shaikh's real competition predicts this outcome for D2-type techniques (lower unit cost, lower profit rate at the going price): the lowest-cost capital cuts the price until its rival's advantage disappears (pp. 314–316), and in Table 7.3 the regulating capitals are the price leaders (row 3) and the lowest-cost condition has the highest sustainable rate under price cutting (row 5, p. 271). The neoclassical and Sraffian view he contrasts with it predicts the opposite selection (C or D1 dominates D2, p. 315). **Agrees with Shaikh.** The controls matter: they sit at 0.48–0.50, so the 0.81 is not a label effect of the single random stream (which moves controls by up to 0.12 elsewhere, results §8.6).

Example: `1_base_mixed_s4` (figure A1). The first dearer order of a profit-rate firm comes at t = 2 (67 dearer orders in the run, `okishio_dearer` 0.62). The unit-cost label's capacity share is 0.81 at t = 50 and 0.93 at t = 100; it takes over (> 90 % of capacity to the end) at t = 93. Over the run the unit-cost firms place 691 orders against 108, 111 of the 124 entrants inherit the unit-cost rule, and the realised rates over t = 100–199 are 14.2 % against 3.0 %. The decomposition over the whole run credits the unit-cost label with +0.27 of sales share through prices, and +0.19 accumulation, +0.18 entry, +0.12 exits on the capacity side. The price is not monotone: 99 at t = 0, 62 at t =

100, a spike to 211 at t = 150 when a shortage follows a wave of scrapping (the excess-demand step of the real competition rule), 78 at t = 199; `price_to_pp` 1.29. Exits are by share from t = 23 on, for both labels, until the profit-rate firms are gone.

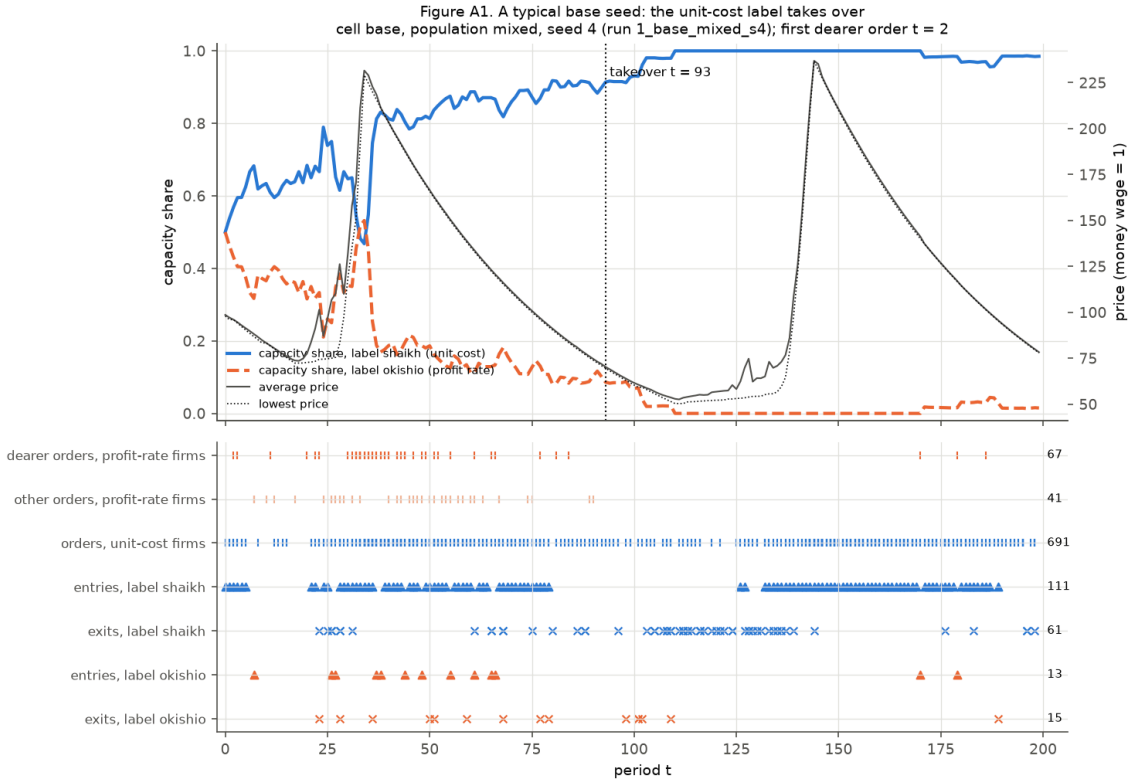


Figure A1 — Seed 4 of `base`: the unit-cost label takes over at t = 93. Top: capacity shares and prices; bottom: the profit-rate firms' dearer orders, other orders, entries and exits.

Example of the other outcome: `1_base_mixed_s1` (figure A2). Seed 1 is one of the 0.19 of `base` seeds the profit-rate rule wins: no takeover, the unit-cost label holds 0.49 of capacity at t = 100 and 0.17 at t = 199. Here the profit-rate firms order dearer techniques less often (`okishio_dearer` 0.38, against 0.51 on average), 59 of 89 entrants inherit the profit-rate rule once it has the majority, and the decomposition runs against the unit-cost label on every term (prices −0.31, accumulation −0.14, exits −0.20). The realised rates are close (9.8 % against 10.6 %). What decides it is who leads when the churn of entry starts: inheritance by market share makes entry a copying process that amplifies the early majority (A.9).

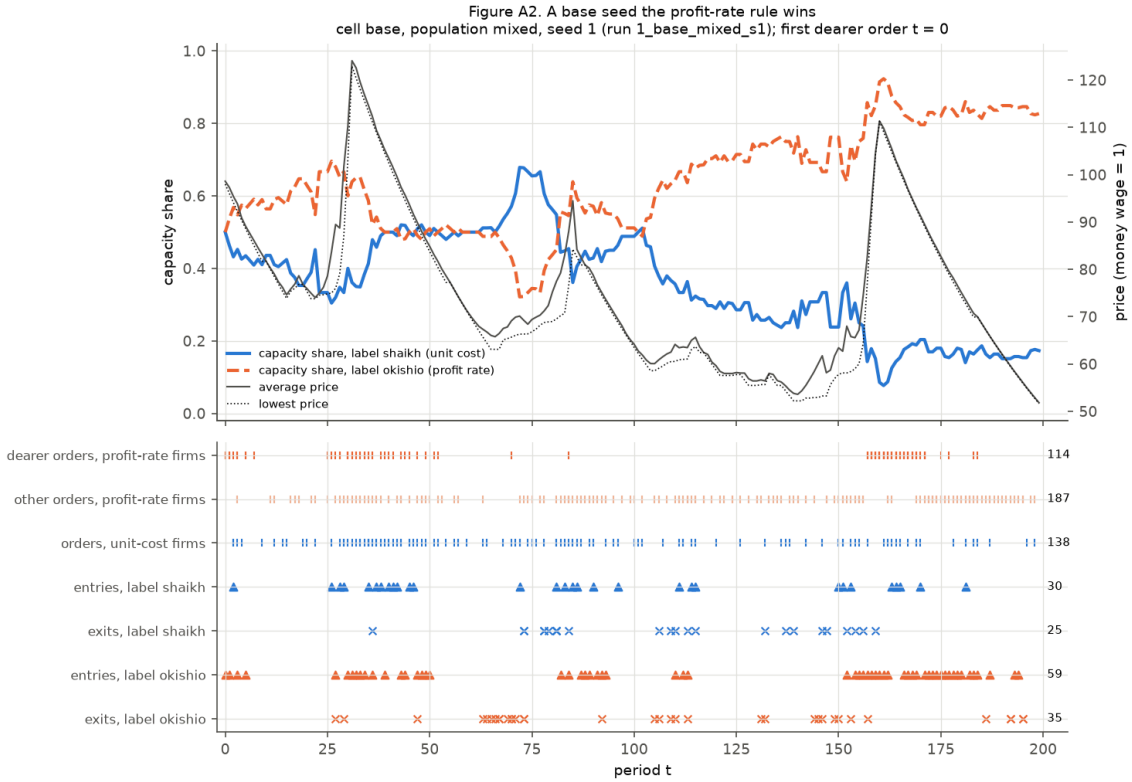


Figure A2 — Seed 1 of `base`, won by the profit-rate rule.

Figure A3 shows a seed of the leader-follow arm at the defaults (`1_lf_mixed_s1`): the unit-cost label takes over at t = 112.

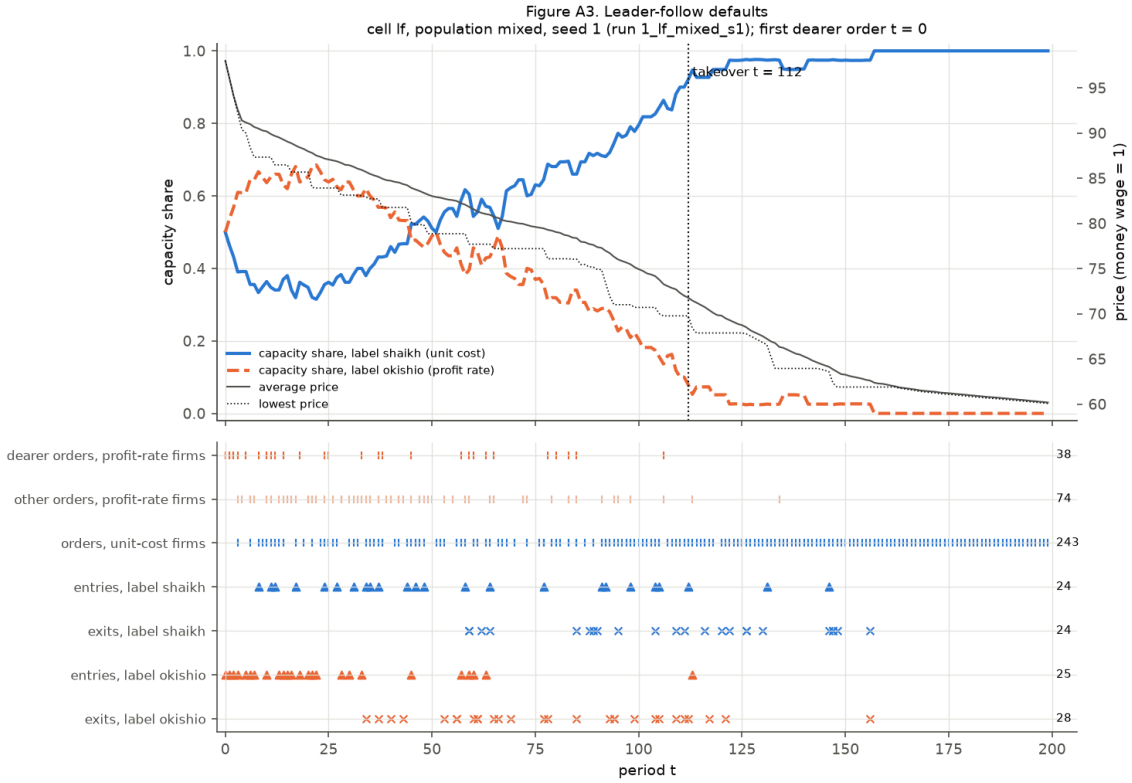


Figure A3 — The leader-follow arm at the defaults (seed 1 of `lf`): the unit-cost label takes over at t = 112.

A.2 No technical change, no contest

Result. `no_tech` (`TECH_DRAWS = False`, fixed menu C, D1, D2) [1, mixed, 1–200]: 0.51 (0.44–0.58), controls 0.45 / 0.45; no takeover in any seed; no entry in the window (`r_new_mean` −0.01); profit rates 6.2 % against 5.9 %; `okishio_dearer` 0.10.

Literature. Shaikh's own figure explains the drift. The profit rate of each technique is linear in the price (eq. 7.11), and below $85 D2 has the highest profit rate of the three (p. 317, fig. 7.22). On the fixed menu the price falls below that point early (mean price over t = 50–199: 77.0 [1, mixed, 1–200]), after which the profit-rate rule picks D2 exactly as the unit-cost rule does: the profit-rate label's mean hurdle is 0.013 and only 0.10 of its orders are not the cheapest technique, almost all of them before the price falls. The two rules become one rule and drift follows. Shaikh's point that the lower-cost method dominates "in real cost-cutting competition" (p. 317) holds here in the weakest sense: the profit-rate firms end up choosing it too. **Agrees with Shaikh, and shows why a fixed menu cannot stage the contest**: the conflict has to be reopened at the current price by a *stream* of new designs, which is how Shaikh (pp. 322–324) and the Duménil–Lévy–Foley framework he builds on pose it (section (b) below).

Example. None needed: no seed has a takeover. The menu is the test industry, not the model.

A.3 Categories against the innovation set

Result. [1, mixed, 1–200]: categories q = 0.5 (`base`) 0.81, q = 0 (`cat_q0`) 0.62; innovation set marx 0.5 / 1.5 / 3 0.80 / 0.94 / 0.97, neutral 0.78. In stage 2 the categories rule averages 0.67 (10 of 24 cells beyond drift) against 0.76–0.91 for the innovation sets. The realised conflict share is 0.50 under categories (an input), 0.14–0.19 under the marx sets and 0.04 under the neutral set (outcomes), yet the innovation set gives the larger advantage.

Literature. Row 6. The conflict between the criteria needs Marx-biased change: with neutral innovation possibilities real competition yields a rising profit rate, as the Okishio theorem does (Shaikh 2016, pp. 323–324, fig. 7.23); with a negative correlation between cost and capital margins (his Table 7.9, Compustat) the criteria part (fig. 7.24). Michl (2023, pp. 47–48) states the same condition in wage–profit terms: the criteria conflict when the new technique raises the maximum profit margin but its wage–profit curve crosses the old one (a switchpoint), and for wages between the two switchpoints Shaikh's criterion adopts techniques that lower the profit rate at the existing wage. **Agrees**, with one twist the literature does not predict: the size of the advantage is not proportional to the share of conflict draws. Under the categories rule half the draws are conflict draws by construction, yet the advantage is smaller (0.81) than under the marx 1.5 set (0.94, conflict share 0.19). What carries the advantage is the profit-rate firms' *orders* of dearer techniques (A.5): under the innovation set a profit-rate firm compares a design with its best known technique at its own price, and with capital-using designs its best technique is often an older, less capital-intensive one, so it keeps ordering a technique that is not the cheapest known (`okishio_dearer` 0.79 under marx 1.5 against 0.51 under categories).

Example: `2_rc-re-static-is15-r16_mixed_s1` (figure A4). Real competition, retained earnings, static demand, innovation set marx 1.5, rbar 0.16. 186 of the profit-rate firms' 215 orders are dearer than the cheapest known technique (`okishio_dearer` 0.87, first at t = 3, last at t = 199); the unit-cost label holds 0.51 at t = 50, 0.75 at t = 100 and takes over at t = 170, through prices (+0.47) and exits (+0.40), not accumulation (+0.01). The price ends at 2.1 times the price of production of the cheapest technique in use, after a shortage: the price signature of A.11.

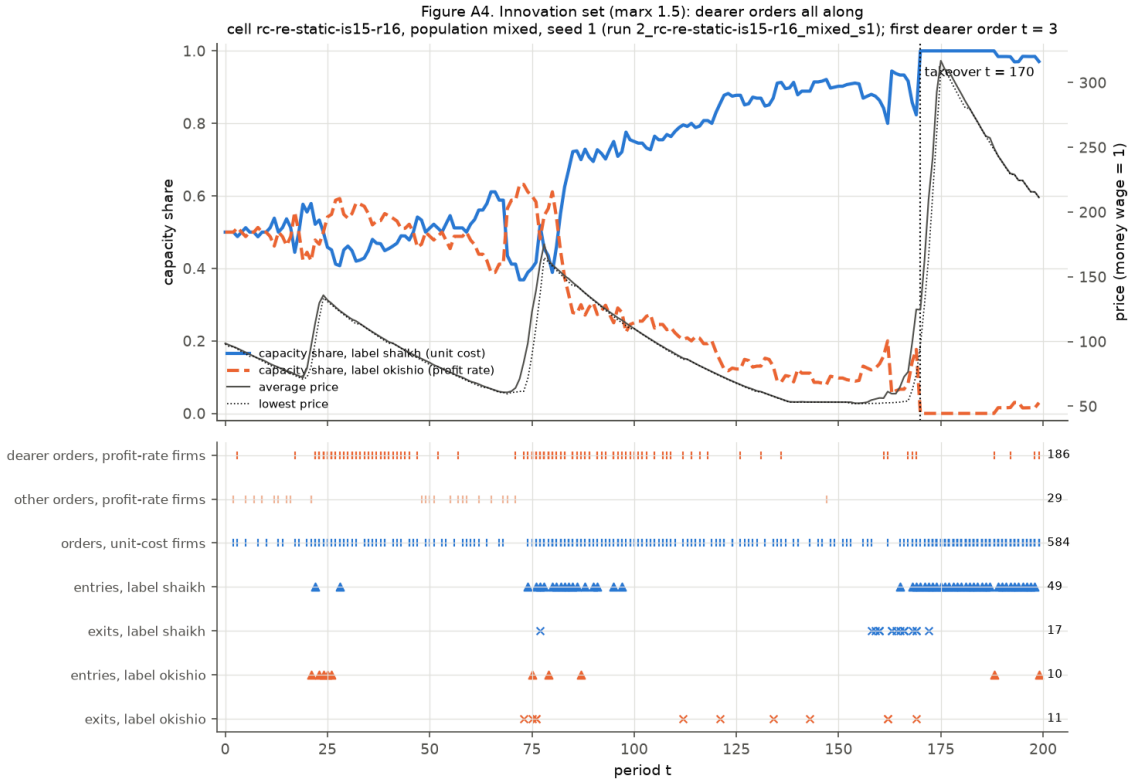


Figure A4 — Innovation set (marx 1.5): dearer orders throughout the run (seed 1 of `rc-re-static-is15-r16`).

A.4 The channel: prices, then exits, not accumulation

Result. Over the 212 mixed cells of stages 1 and 2 the whole-run price term of the unit-cost label's sales share correlates 0.80 with the share won and the accumulation term 0.60; the price term is the larger in 93 % of cells (results, §5). In `base` [3, mixed, 1–500] the unit-cost label gains +0.25 of sales share through prices and +0.28 of capacity share through the exits of profit-rate firms, while accumulation nets to −0.003 and entry to +0.03 (figure B1.2).

Literature. Row 1 and row 7. Shaikh's mechanism runs through the price: the cost-cutter "makes room for itself" by cutting (p. 314) and the rival's market share and then its plants go (p. 316). Nelson and Winter's selection runs through investment keyed to profitability (pp. 17, 19), and so does Duménil and Lévy's (1995, p. 217: profitability generates the cash flow that finances growth). **The model sides with Shaikh on the channel**: the profit-rate firms do not accumulate faster at the defaults, because they do not realise the higher rate (A.6), and because idle scrapping (`IDLE_SCRAPPING`, two idle periods) destroys capacity that sales do not use, so accumulation without customers does not stick. The exits are the second half of Shaikh's story (the old regulating capital "is doomed", p. 314): in `base` they carry more capacity share than any other term.

Example. `1_base_mixed_s4` (A.1): prices +0.27, exits +0.12, but also accumulation +0.19 and entry +0.18 once the unit-cost label leads: the channels compound after the takeover.

A.5 The mechanism: the profit-rate rule's switch to dearer techniques

Result. `okishio_dearer` (share of profit-rate orders that are not the cheapest known technique) correlates 0.77 with the share won across cells, the strongest of all observables; 0.49 in stage 2 seeds won by the unit-cost rule against 0.17 in seeds it loses; 0.51 in `base`, 0.79–0.85 under the marx sets, 0.10 on the fixed menu. Within cells (B1, cell fixed effects, 17 585 decided stage 2 seeds) a one-standard-deviation increase in `okishio_dearer` (0.18) raises the probability that the unit-cost rule wins by 0.11 (coefficient 0.64, 95 % interval 0.56–0.71).

Literature. This is Shaikh's case in Tables 7.10–7.12: the profit-rate rule picks the technique with the higher rate at the ruling price ($100), the lower-cost one drives the price down to where the ranking of profit rates reverses (below $85 in fig. 7.22, p. 317), and from then on the dearer choice is a handicap. Okishio's rule is not irrational at

the price it sees; it is evaluated at a price that the cost-cutters' adoption will lower. **Agrees.** It also matches Duménil and Lévy's point (2005, p. 209) that the criterion itself is classical; what decides is the price at which it is applied (row 6, and the B3 candidate on the evaluation price).

Example. Compare the two `base` seeds: 67 dearer orders in seed 4 (unit-cost takeover at t = 93) and an `okishio_dearer` of 0.38 in seed 1 (profit-rate win); in the anchor and leader-follow reversals (A.7) the profit-rate firms almost never order dearer techniques (0.06–0.10), so the mechanism is absent and something else decides.

A.6 Realised profit rates and prices

Result. The unit-cost firms earn the higher realised profit rate: `base` [3, mixed, 1–500] 8.4 % against 4.7 % (results: 8.4 % against 4.5 % in stage 1); across stage 2 decided seeds 11.1 % against 8.8 % where the unit-cost rule wins, and 7.8 % against 9.9 % where the profit-rate rule wins (13 956 and 3 644 seeds): **the winner earns more, whichever rule it uses**. Seeds won by the unit-cost rule have *higher* mean prices (74.5 against 62.3) and end at the price of production of the cheapest running technique (`price_to_pp` 1.13 under real competition) rather than below it (0.87). The leader is the lowest-cost firm in 0.94 of periods (`base`).

A result the literature frames but the results document did not state: selection picks the rule that lowers the industry's profit rate. Compare the two neutral controls of the same cell, i.e. an industry in which every firm uses one rule (`profit_rate_total`, mean over t = 100–199, 95 % interval of the mean; [stage, control, seeds]):

Cell	All unit cost	All profit rate	Mixed	Share won (mixed)
`base` (categories) [3, 1–500]	8.72 % (8.50–8.94)	9.03 % (8.81–9.26)	8.85 %	0.81
`is_neutral` [1, 1–200]	8.58 % (8.34–8.83)	9.60 % (9.15–10.04)	8.85 %	0.78
`is_marx_15` [1, 1–200]	8.50 % (8.21–8.79)	10.91 % (10.48–11.35)	9.18 %	0.94
`is_marx_3` [1, 1–200]	7.87 % (7.56–8.18)	10.81 % (10.39–11.23)	8.49 %	0.97
`rc-re-static-is3-r16` [3, 1–500]	12.03 % (11.59–12.47)	18.67 % (18.26–19.07)	13.70 %	0.99

Under the Marx-biased innovation sets an industry run on the unit-cost rule realises a profit rate 2.4 to 6.6 points lower than the same industry run on the profit-rate rule, at similar prices (mean price over t = 50–199: 76.8 against 74.9 under marx 1.5) and similar unit costs at the end (51.7 against 50.4): the difference is capital per unit of output. Yet in the mixed industry the unit-cost rule is selected in 0.94–0.99 of seeds. This is Shaikh's claim of p. 261 and pp. 320–322 at the scale of one industry (the cost-cutting criterion forced by competition lowers the rate of profit relative to the profit-rate criterion, at a given real wage) and it is the prisoner's dilemma that Foley (1986, p. 131, cited by Michl 2023, p. 46) and Baldani and Michl (2000, *scanned, not read*) see in Marx: each capitalist is forced to adopt what lowers the common rate. Under the categories rule the difference is small and the intervals touch, because half the designs are adopted by both rules. **Figure A8** shows the table with its intervals. On the fixed menu the ordering is reversed (`no_tech` [1, controls, 1–200]: 8.81 % all unit cost against 5.75 % all profit rate), but for a reason that has nothing to do with capital intensity: when every firm builds D2, no firm keeps a cost edge, the real-competition rule never cuts, and the price stays high (mean 94.1 over t = 50–199, against 76.0).

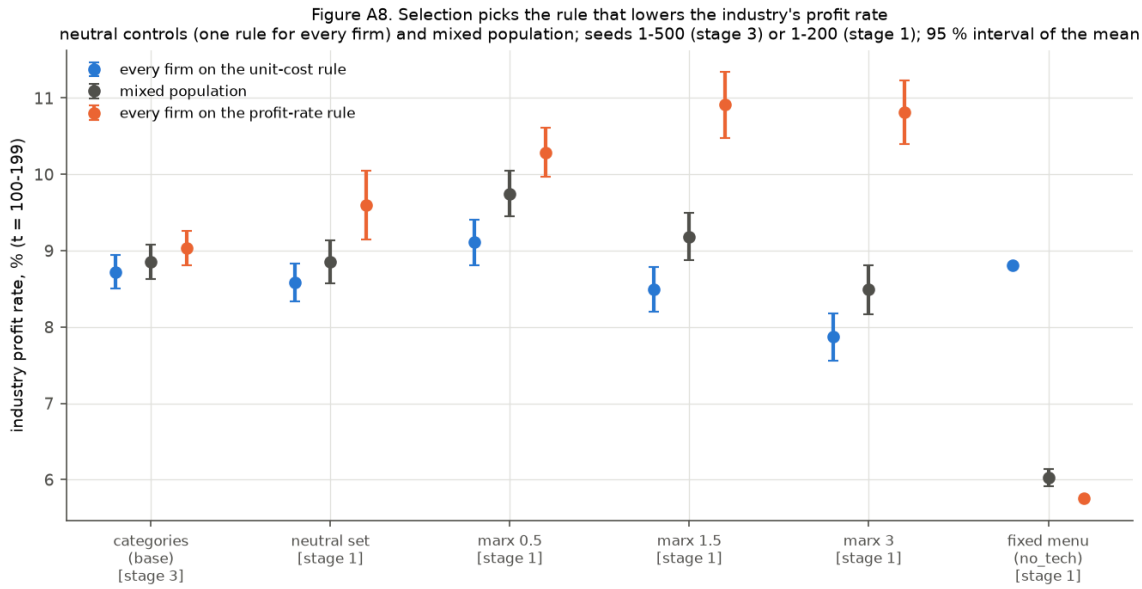


Figure A8 — Industry profit rate in the neutral controls (one rule for every firm) and in the mixed population, by technical-change rule; 95 % intervals of the mean.

(a) Why does the profit-rate rule not realise the higher profit rate it maximises at the current price? Two hypotheses, which Part B (B1) separates with the stage 2 seeds:

1. *Shaikh's argument.* The rate the profit-rate firm maximises is computed at its own current price; the cost-cutters' adoptions then lower the price, and at the lower price the dearer technique earns less than the cheaper one (fig. 7.22: D2 overtakes D1 below $85, p. 317). The firm locks capital into a technique whose rate is never earned. This predicts that the realised gap grows with the share of dearer orders, *among seeds with the same winner*.
2. *Market structure.* The rate is a price leader's premium: whoever wins leads a more concentrated market (HHI 0.24 where the unit-cost rule wins, 0.34 where it loses, stage 2) and posts the higher price; the rule does not matter once the winner is known. This predicts that the gap goes with the winner and with concentration, not with `okishio_dearer` once the winner is controlled.

B1's answer, in one line: **both, and Shaikh's mechanism survives the control** (gap on `okishio_dearer` 0.051 per unit, 95 % interval 0.038–0.065, holding the winner and concentration fixed within cells; the winner's own premium is 0.038, 0.029–0.047; concentration adds little, 0.017 per unit of HHI, 0.002–0.032).

For the presentation, the group can say: the profit-rate rule maximises the rate at a price that its rivals' choices will not let it keep; the realised rate then goes to the cost-cutters, and on top of that to whichever firm leads. The model reproduces Shaikh's mechanism; it does not reproduce a rule that is irrational at the price it sees.

A.7 The two reversals

The profit-rate rule wins beyond drift in two configurations only.

(i) **`anchor_r16`** (version 1 anchor pricing: each firm moves toward its own target-return price uc + rbar κ, wars after low utilisation; rbar 0.16) [1, mixed, 1–200]: 0.26 (0.20–0.33), 13 undecided seeds, controls 0.53 / 0.51. At rbar 0.05 (`anchor_r05`) 0.66, at drift. In the reversal: 0.05 entries in the window, two firms at the end (`n_firms_end` 2.04), HHI 0.74, both channels against the unit-cost label (prices −0.18, accumulation −0.21 over the whole run), `okishio_dearer` 0.10, `price_to_pp` 0.54.

Literature. The pricing brief named this trap before anything was run (`mechanism-briefs.md`, "The trap: the anchor price decides the winner"): a target-return anchor makes the technique with the lowest uc + r̄ κ the cheapest in normal times, which at r̄ = 0.16 is C or D1, not D2 (C 100, D1 98.8, D2 100.3; at r̄ = 0.05: D2 82.9, D1 83.1, C 84.9). That ranking is Sraffa's criterion of choice (the lower price of production at the ruling rate, Shaikh 2016, p. 320 n. 36), and it is the profit-rate rule's criterion with h = r̄. In other words the anchor rule builds the Okishio criterion into the price, which is the situation Duménil and Lévy (2005, p. 209) and Okishio's theorem presuppose:

prices of production prevail. Shaikh's answer to that situation (p. 321) is that actual choices are made at actual and projected market prices, which differ from prices of production. That was the first reading, written before Part B. **Part B refutes it: the trap is not what reverses the outcome.** With one anchor for all firms (the leader's target-return price) the profit-rate rule still wins (0.32), with the *markup* anchor, which favours the lowest unit cost by construction, it still wins (0.27), and with the accelerator instead of retained earnings the reversal disappears (0.47, at drift); the reversal sets in between rbar 0.08 and 0.10, as entry closes (B2). What the anchor runs show (B1) is a closed industry in which growth is financed out of profits: no entrant arrives (the realised rate on new plants, 5–7 %, never reaches 16 %), the unit-cost firm piles up capital-intensive designs (capital per unit 160 → 233 → 359 over t = 1–25, 26–75, 76–125 in `1_anchor_r16_mixed_s3`, against 149 → 167 → 204), so that each unit of retained profit buys it ever less capacity (capacity added = retained profit / κ of the new plant), its old plants are retired in a war, and the minimum of two firms keeps a duopoly. Smaller plants make the reversal stronger, not weaker (0.08 with plants of 10, B2): it is capital per unit, not the indivisibility of plants, that slows the unit-cost firm. And the retirement in wars is the decisive step: with the version 1 scrapping test (operating cost, Salter's margin) the same cell gives the unit-cost rule 0.90 (B2). **The first reversal is a property of the version 2 scrapping test in a closed industry, not of real competition against Okishio.**

Example: `1_anchor_r16_mixed_s3` *(figure A5).* First dearer order at t = 1 (17 in all); the unit-cost label holds 0.54 of capacity at t = 50 and 0.42 at t = 100; four exits, all by non-renewal (o16, s59, s98, o127); no entry; at t = 129 the last unit-cost plants are retired and the label's capacity falls to zero; the firm stays as one of the two firms `MIN_FIRMS` keeps. Price 100 → 71 (t = 50) → 68 (t = 100) → 51 (t = 199).

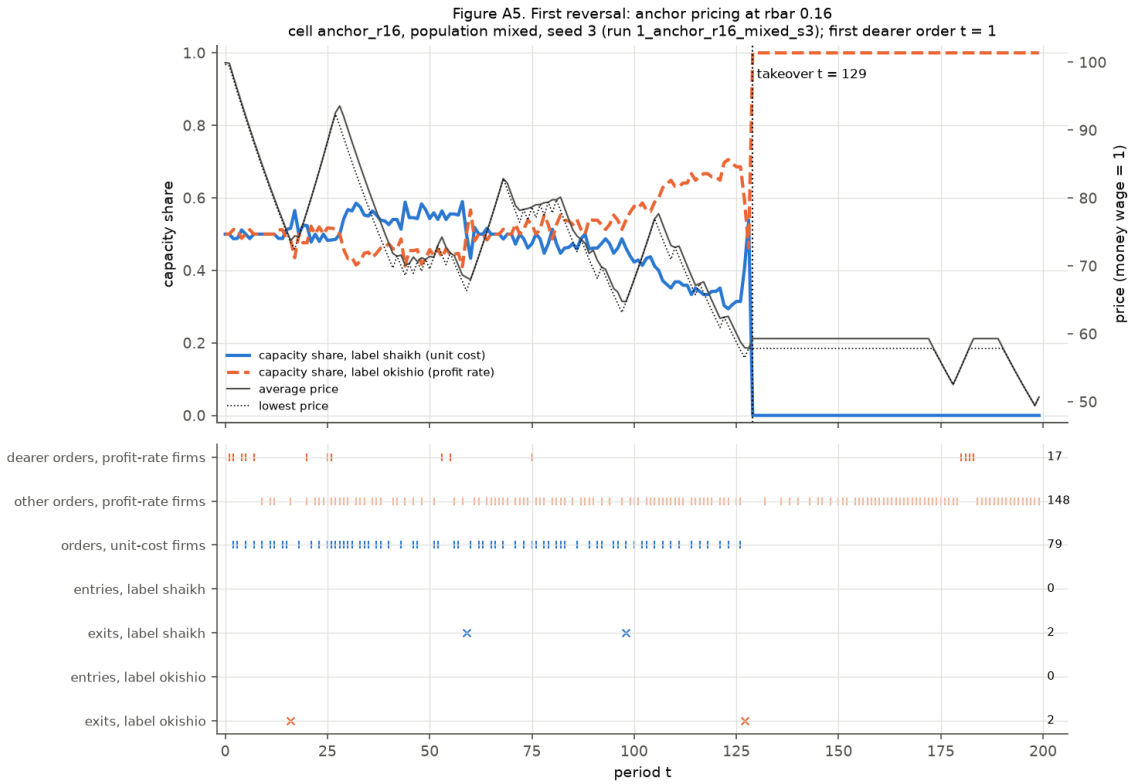


Figure A5 — First reversal (anchor pricing, rbar 0.16): the unit-cost label's capacity falls to zero at t = 129 (seed 3 of `anchor_r16`).

(ii) **`lf-re-g01-cat-r16`** (leader-follow, retained earnings, demand growing 1 %, categories, rbar 0.16) [3, mixed, 1–500]: 0.32 (0.28–0.36), controls 0.49 / 0.49; its finance twin `lf-acd-g01-cat-r16` (accelerator with borrowing) [3, mixed, 1–500]: **0.85** (0.82–0.88). In the reversal: 0.24 entries in the window, `n_firms_end` 2.01, HHI 0.91, profit-rate firms earning more (6.3 % against 3.3 %), `okishio_dearer` 0.06, the unit-cost label's capital older (32.6 against 22.7 periods), takeovers early (median t = 82; 0.75 of them before t = 100). In the twin: 7.0 entries, 9.4 firms, both channels for the unit-cost label (+0.32 prices, +0.31 accumulation).

Literature. Row 7. This is Nelson and Winter's selection through investment keyed to profitability (pp. 17, 19) and Duménil and Lévy's (1995, p. 217) profitability-financed growth, in the one corner where it can work: demand grows, so capacity rather than price is the constraint; investment is self-financed, so capacity grows with retained profits, and at a common price the technique with the higher profit rate yields more capacity per unit of profit (units = retained earnings / κ); no entrant arrives at rbar 0.16 to carry the other rule; and under the categories rule the unit-cost firms build every conflict-zone design, whose unit-cost saving is small and whose extra capital is large, while leader-follow pricing does not reward a small cost edge (every firm cuts to its floor whenever capacity is not short). Under the accelerator, investment follows sales instead of profits, and sales follow price: the price channel drives accumulation and the unit-cost rule wins. **The investment rule decides the contest here, which is the second half of the research question; Part B adds that borrowing plays no part in it** (not one of the 500 seeds of the twin carries debt: the accelerator is financed out of cash). It does not contradict Shaikh, whose price-cutting argument needs a firm that can expand capacity to take the market (p. 314: the cost-cutter "makes room for itself"); with retained earnings, thin margins and capital-heavy plants the unit-cost firm cannot.

Example: `3_lf-re-g01-cat-r16_mixed_s1` *against* `3_lf-acd-g01-cat-r16_mixed_s1` *(figures A6, A7; same seed).* Retained earnings: 5 dearer orders, all before t = 13; the unit-cost label holds 0.55 at t = 50 and 0.46 at t = 100, then loses steadily as the profit-rate firms place 765 orders against 178 (the unit-cost firms' last order is at t ≈ 145); takeover by the profit-rate label at t = 132; four exits by share (o54, o76, s80, s93), no entry; the price falls every period, 98 → 78 → 62 → 48; capital age of the unit-cost label 54 periods at t = 199 against 14. Accelerator with borrowing: 2 dearer orders; the profit-rate firms place only 10 orders in the whole run (they lose sales, so the accelerator does not ask them to invest), the unit-cost firms 339; takeover by the unit-cost label at t = 73; 8 entrants, all inheriting the unit-cost rule; the price falls to about 75 by t = 75 and then rises to 103–118 under recurring shortages.

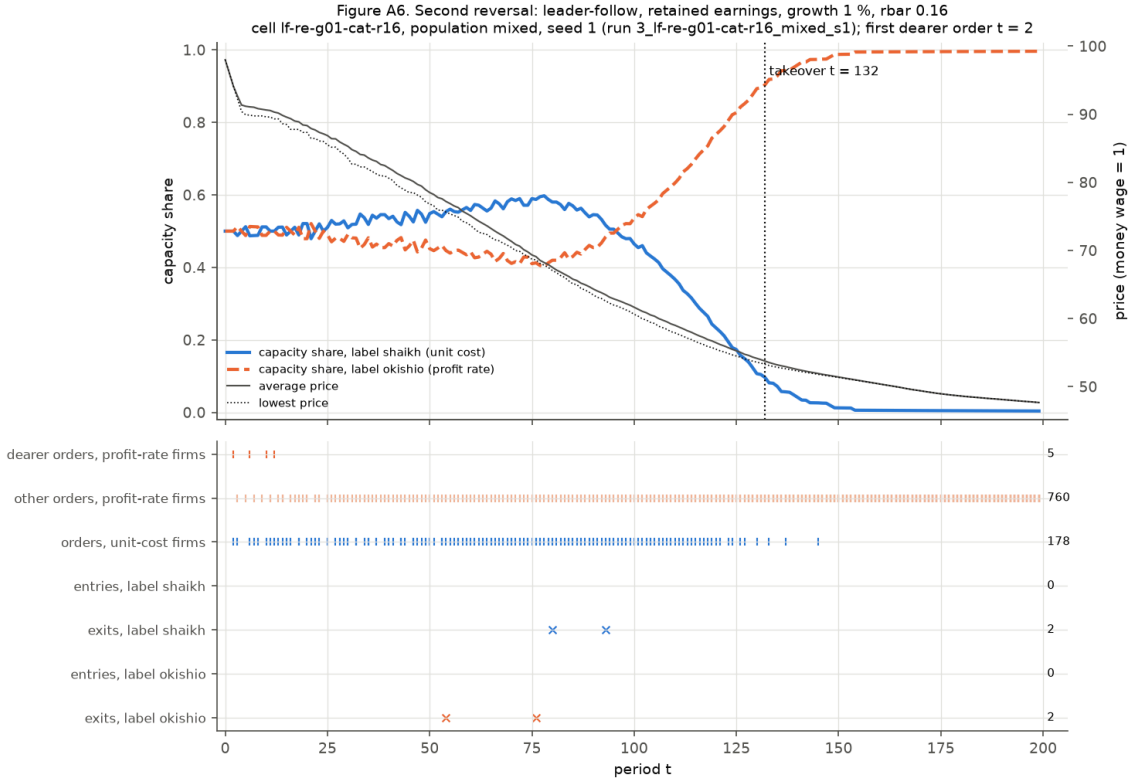


Figure A6 — Second reversal (leader-follow, retained earnings, growing demand): the profit-rate label takes over at t = 132 (seed 1 of `lf-re-g01-cat-r16`).

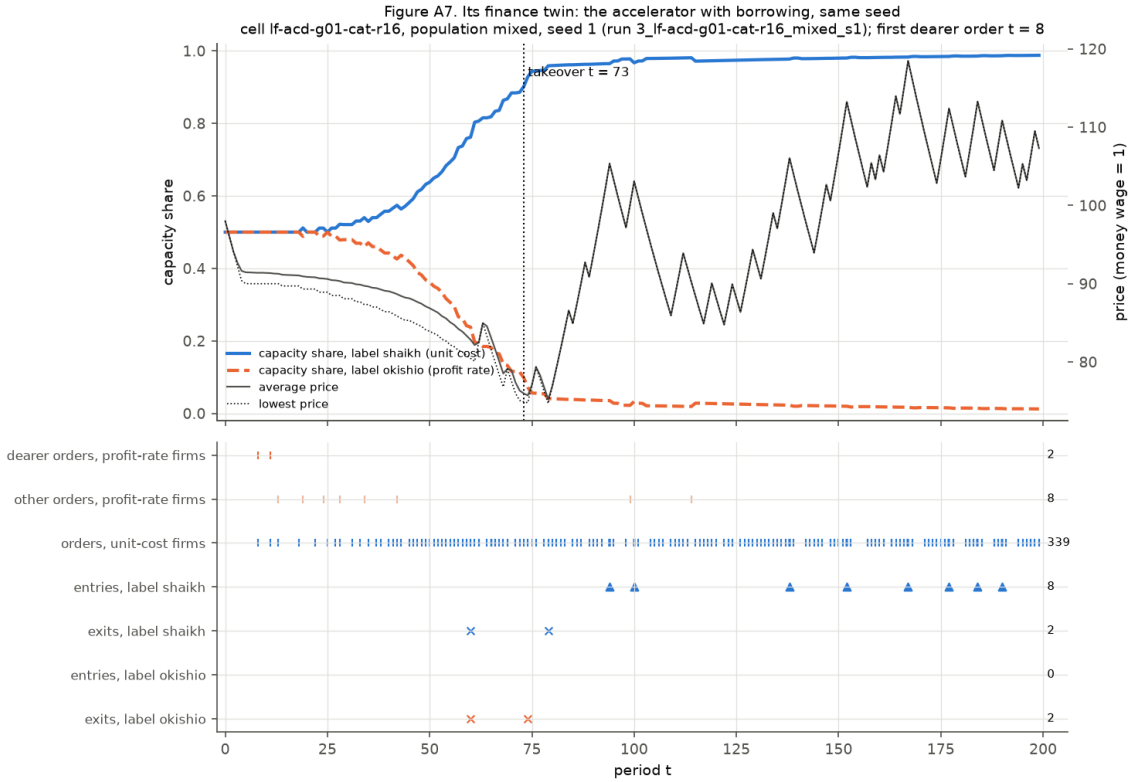


Figure A7 — The same seed with the accelerator and borrowing (`lf-acd-g01-cat-r16`): the unit-cost label takes over at t = 73.

A.8 The factors that move nothing

Result. CHI 0.25 / 0.5 / 2 / 4: 0.71 / 0.79 / 0.75 / 0.79 [1, mixed, 1–200], none beyond `base`'s interval; logit with η = 1 identical to the replicator; floor of the default rule (0.74); excess-demand signal (0.76); entry speed θ = 1 (0.79); scrapping 4 / 4 (0.78); depreciation by use (0.75); plant size 40 (0.79); τ = 0.25 (0.78); the hurdle floor (inert); the leader-follow shortage branch (inert).

Literature: the χ trap. The demand brief warned that weak selection would let the firms with higher profit rates win by accumulating faster, and that selection is weak in the data (Dosi et al. 2015; Foster, Haltiwanger & Syverson 2008, p. 397, where demand differences, not physical productivity, dominate survival). The trap does not spring, for three reasons visible in the runs: (1) the profit-rate firms do not have the higher realised rate to accumulate with (A.6), so a slow market does not hand them anything; (2) idle plants are scrapped after two idle periods, so capacity that sales do not use disappears; (3) χ sets the *speed* of share movements, not their direction, and the share won is read at t = 199: at χ = 0.25 the mean takeover comes at t = 174 instead of 117, and fewer seeds have finished (cap share 0.63), but the same rule leads. In the phase block χ interacts with finance (results §3): with borrowing under real competition a *higher* χ hurts the unit-cost rule. **The briefs' trap assumed that the Okishio firm earns the rate it maximises; the model says it does not, which disarms the trap.**

τ: the briefs note that on the fixed menu τ above about 0.12 makes D2 stop being a conflict case against C (at p = 100, r_D2 = 25 / (157.9 + 59.2 τ) exceeds r_C = 22 / (137.5 + 64.25 τ) for τ > 0.119), while `tau_025` still gives 0.78. Drawn designs keep a conflict zone because for a *local* design the profit-rate rule adopts iff b < (1 + h τ) a ucs_o / ((δ + h) κ_o) while the unit-cost rule adopts iff b < a ucs_o / (δ κ_o): the zone between exists whenever h > 0 and δ τ < 1, i.e. always (τ only widens the profit-rate rule's acceptance by the factor 1 + h τ, about 1.02 at h = 0.09). Under the categories rule the placement of conflict draws (`draw_design`) does not even read τ, so q = 0.5 of draws still fall where only the unit-cost rule adopts. The menu loses its conflict because D2 against C is one large, discrete step; a stream of small Marx-biased steps keeps one. A τ sweep is therefore predictable on paper and was given the lowest priority in Part B.

A.9 Demand met and turnover

Result. Unmet demand is below 3 % of demand except under the anchor rule (`anchor_r16` 7.6 %, `anchor_r05` 10.3 %) and the leader-follow operating-cost floor (25.7 %, where the industry collapses in 81 of 200 seeds); it does not move with the winner (correlation 0.06). Turnover: more entry and more firms where the unit-cost rule wins (19.9 entries in the window against 13.3; HHI 0.24 against 0.34, stage 2); bankruptcies only with borrowing (0.6 against 0.3 per seed).

Literature. Row 10. Shaikh's entry equalises the profit rate on new investment (p. 296); here `r_new_mean` is 0.054 against rbar 0.05 in `base` and the industry rate is above rbar in 95 of 100 periods, so the trigger fires almost every period (review-v2 §2.2, point 5, called it churn). Entrants copy the rule of an incumbent drawn by market share (`equations/inherited_rule.py`), so entry is a copying process that carries whichever rule has the majority: that is why entry is Shaikh's long-run carrier only *after* the price channel has made the unit-cost rule the majority, and why it amplified the profit-rate majority in `1_base_mixed_s1`. At rbar 0.16 the industry rate is above rbar in 0.3–2.3 of 100 periods, entry stops, and the two reversals both sit there. **Agrees with Shaikh on entry as carrier; adds that the carrier is neutral (it copies the majority).**

A.10 Marx bias as an outcome

Result. Realised change is Marx-biased ($\gamma > 0$ and $\chi > 0$ for the label's plants between t = 100 and t = 199) for the unit-cost label in 0.84 of `base` seeds against 0.29 for the profit-rate label; under the neutral set for neither (0.00 / 0.005: capital-saving designs dominate); under marx 1.5 1.00 / 0.71.

Literature. Michl (2023, pp. 37–41) reviews the evidence of Marx-biased technical change (falling capital productivity with rising labour productivity, the "fossil production function"); Shaikh (2016, p. 324) reads the same pattern in Compustat data (Table 7.9). Duménil and Lévy (1995, p. 214) assume neutral, local innovation and obtain the bias from selection under a rising labour cost. **The model cannot reproduce Duménil and Lévy's route: the money wage is the numéraire, so labour never gets dearer, and a neutral set selected at a fixed wage produces capital-saving change.** With a fixed wage, Marx bias must be assumed in the innovation set; it is then reproduced as an outcome mostly by the unit-cost label.

A.11 The price signature

Result. Under real competition the price at t = 199 exceeds the price at t = 100 in 0.17 (marx 0.5) to 0.31 (marx 3) of seeds and ends 10–20 % above the price of production of the cheapest technique in use; under leader-follow at most 0.06, ending below it (0.85–0.93) (figure 5 of the results). `base` [3, mixed, 1–500]: 0.23 of seeds, `price_to_pp` 1.05.

Literature. Row 3. Shaikh (p. 316) predicts the D2 price path exactly: the price falls while the new regulating capital shoulders the old one aside, then rises to the new regulating price of production, which for a capital-using technique is *higher* than the old one ($101.30 against $100); he notes that this rise is what the neoclassical and Sraffian approach rules out (p. 316). **Agrees under real competition**, where the rise comes through the shortage step (prices move up to the next rival's price or by 0.5 × the excess-demand share), as in `1_base_mixed_s4` (spike to 211 at t = 150) and `2_rc-re-static-is15-r16_mixed_s1` (2.1 × the price of production at t = 199). Leader-follow matches every price to the leader and cuts every period, so the rise has no room. The rise in the model comes from shortages after scrapping, not from entry pricing at the normal rate as in Shaikh's account (entry at rbar 0.05 is permanent, so it cannot pull the price up to a target).

(b) The technical-change rule is the model, not a detail

`no_tech` at drift (A.2) means the research question is answered by the technical-change rule. What each rule assumes about the direction of innovation:

- **Categories** (`equations/new_technique.py`): every design lowers the unit cost of the firm's reference technique (ucs falls by a ≤ 5 %, κ rises by b with b < b_S), so the unit-cost rule adopts every design; a coin flip (q = 0.5) puts the design in the conflict zone b_O ≤ b < b_S, computed at the firm's own price. Innovation is

cost-reducing by construction, Marx-biased by construction, and the conflict share is an input (`review-v2.md` §4.1, points 1–4).

- **Innovation set, marx** (`equations/innovation_set.py`): ucs falls by a, κ rises by k a with k log-normal around TECH_BIAS; some designs raise unit cost and are rejected by both rules; zones are outcomes of the draw and of the price. Marx-biased by construction, cost-reducing or not by draw.
- **Innovation set, neutral**: direction uniform on the circle; Marx bias is not assumed.

Which does the literature support? Shaikh (pp. 322–324) frames the question as one about the *probabilities* of the two kinds of technical change and argues, with Marx, that cost-reduction typically comes with more capital per unit (negative correlation in Table 7.9); Michl (2023, pp. 37–41) and Foley (as reported by Michl) treat Marx-biased technical change as the empirical pattern; Duménil and Lévy (1995, p. 214) start from neutral, local innovation and obtain the bias through rising wages. With a fixed wage the model can only take the first route, so **the marx innovation set is the specification the literature supports for this model, the neutral set is its natural robustness check, and the categories rule is a stylised version of the marx set with the conflict share fixed at q** (its advantage is that q is a transparent treatment). The group should present the technical-change rule as the model's core assumption, with `no_tech` as the null.

3. Part B: going further

What was run, in one table (all with both neutral controls over the same seeds; tables in `outputs/analysis/`, stage 4 rows appended to `runs.csv` and `cells.csv` with stage = 4, candidate rows in `outputs/analysis/b3/`):

Track	Item	Seeds	Done
B1	Regression of the realised gap and of `shaikh_wins` (stage 2, 17 585 seeds)	existing	yes
B1	Whole-run attribution for the cells of section 6 (figure B1.2)	existing	yes
B1	Anchor reversal: reruns of seeds 3, 4, 5 of `anchor_r16` and seed 3 of `anchor_r05`	reruns	yes
B1	Finance twin side by side, seed 1 (figure B1.7)	reruns	yes
B1	Takeover timing (figure B1.3, `b1_takeover.csv`)	existing	yes
B2	`rc-re-g01-cat-r16`	1–500	yes
B2	Anchor threshold: rbar 0.08, 0.10, 0.12, 0.14 (0.05 and 0.16 from stage 1)	1–200 (0.10, 0.12: 1–500)	yes
B2	Added by Part A: the leader-follow reversal with the best plant's unit cost as floor; with `MIN_FIRMS` = 1; with the accelerator without borrowing. The anchor reversal with one common anchor, with the markup anchor, with the accelerator, with `MIN_FIRMS` = 1	1–200	yes
B2	Finance × CHI: accelerator with borrowing, CHI 0.25 and 0.5, both pricing rules	1–200	yes
B2	TAU sweep: 0.05, 0.10, 0.15 (categories), 0.10, 0.25 (marx 1.5)	1–200	yes
B2	Added after the first B2 results: plant size 10 and scrapping on operating cost in the reversal cells; scrapping on operating cost at the defaults	1–200	yes
B3	Two candidate rules behind new switches, at the base, leader-follow, rbar 0.16 and marx 1.5 cells	1–200	yes

Seeds were cut from the prompt's 500 to 200 for the B2 cells other than the real-competition twin (later also `anchor_r10`, `anchor_r12` and `scrap_ucs` were taken to 500), as section 6 of the prompt allows: the machine had 4 + 2 cores, not 10 (measured: 17 runs a second on the 4-core machine, 6 on the 2-core one).

B1. The mechanism, on existing data

The realised profit-rate gap (open question 2). OLS over the 17 585 decided mixed seeds of stage 2 with cell fixed effects, standard errors clustered by the 176 cells (`b1_regression.csv`):

Outcome	Specification	`okishio_dearer`	`price_to_pp`	`hhi_mean`	`n_firms_mean`	`war_episodes`	`shaikh_wins`	within R^2
gap	mechanism	**0.075** (0.060, 0.091)	0.010 (0.003, 0.017)	0.013 (−0.005, 0.031)	0.000 (−0.001, 0.001)	0.0011 (0.0006, 0.0016)		0.08
gap	+ winner	**0.051** (0.038, 0.065)	0.011 (0.004, 0.017)	0.017 (0.002, 0.032)	0.000	0.0011 (0.0007, 0.0016)	**0.038** (0.029, 0.047)	0.15
gap	prompt's list (+ decomposition)	0.016 (0.004, 0.029)	0.011 (0.005, 0.018)	0.007 (−0.007, 0.020)	0.000	0.0009 (0.0004, 0.0014)		0.24
`shaikh_wins`	mechanism	**0.635** (0.558, 0.713)	−0.013 (−0.023, −0.004)	−0.098 (−0.237, 0.041)	−0.001	−0.0006		0.09

(gap = realised profit rate of the unit-cost label minus that of the profit-rate label over t = 100–199; in the prompt's list `d_share_price_all` and `d_share_accum_all` enter with 0.068 and 0.057: they are parts of the outcome, which is why the second line is the one to read.) Figure B1.1 plots the gap against `okishio_dearer` with a pooled line for each pricing rule.

Reading. Among seeds of the same cell *and with the same winner*, every 0.1 more of dearer orders by the profit-rate firms widens the gap by half a point of profit rate: **Shaikh's hypothesis holds within cells**. The winner has its own premium of 3.8 points, and concentration adds a little (HHI coefficient 0.017): **the market-structure hypothesis holds too, as a separate premium**. The number of firms adds nothing once HHI is in. The share of dearer orders is also the strongest within-cell predictor of who wins (a 0.18 standard deviation raises the unit-cost rule's chance by 0.11).

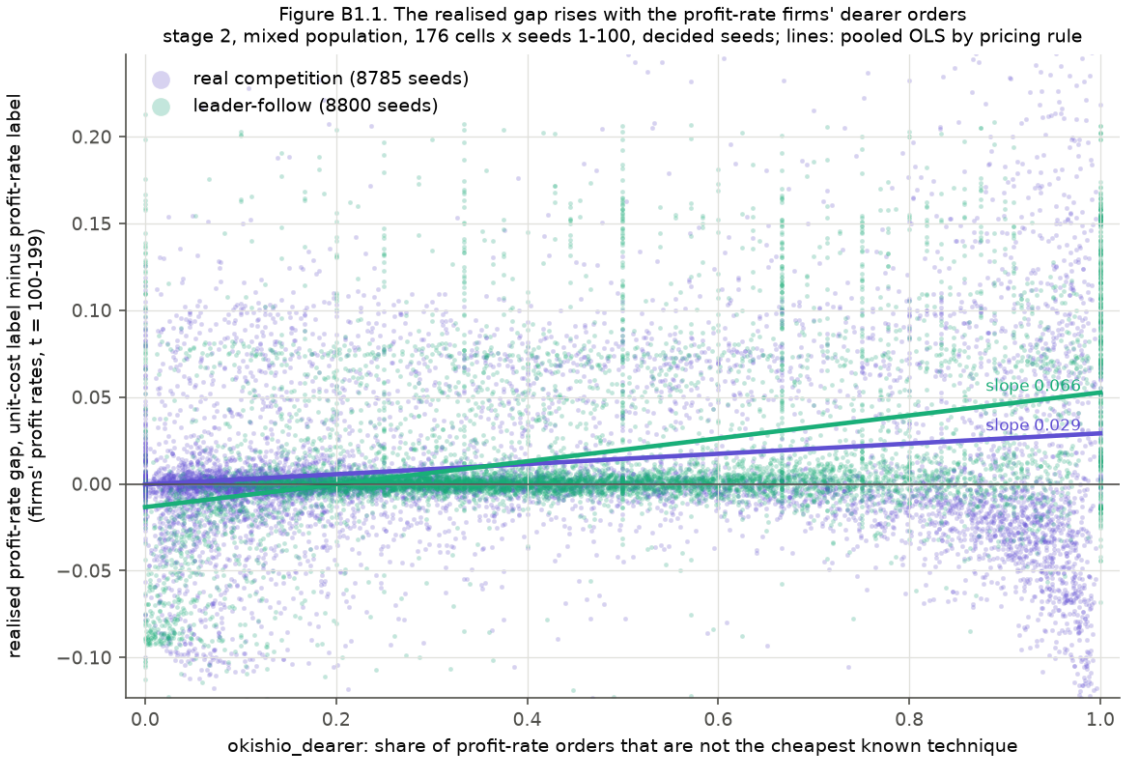


Figure B1.1 — Realised profit-rate gap (unit-cost label minus profit-rate label, t = 100–199) against the profit-rate firms' share of dearer orders, decided seeds of stage 2.

Whole-run attribution (figure B1.2, `b1_attribution.csv`). The cells of section 6 split into three groups. *Price, then exits*: `base`, `is_marx_15` (capacity +0.28 and +0.32 through the exits of profit-rate firms; sales +0.25 and +0.27 through prices). *Accumulation for the unit-cost rule*: `lf` (+0.09), the leader-follow twin with the accelerator (+0.31), the operating-cost floor (+0.32) and `anchor_r05` (+0.20). *Accumulation against it*: `anchor_r16` (−0.21), `lf-re-g01-cat-r16` (−0.18), `rc-re-g01-cat-r16` (−0.20 at 100 seeds). Entry terms are small everywhere (−0.06 to +0.03): entrants copy the majority, so they move little share between labels.

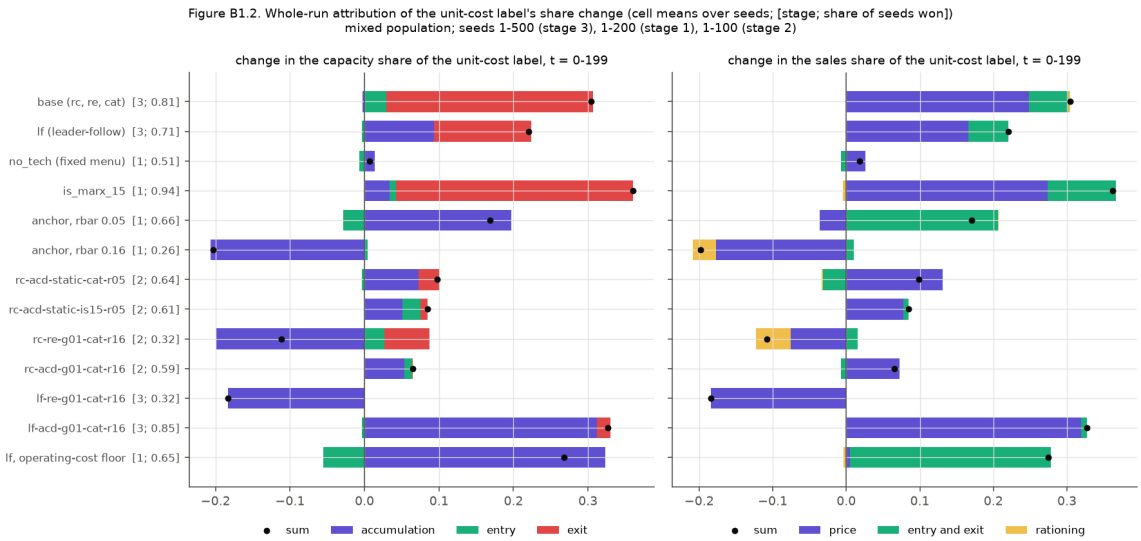


Figure B1.2 — Whole-run decomposition of the unit-cost label's share (price, accumulation, exits, entry) in the cells of section 6.

The anchor reversal (figures A5, B1.4–B1.6; reruns of `1_anchor_r16_mixed_s3`, `_s4`, `_s5` and `1_anchor_r05_mixed_s3`).

- *When the anchor sets the price.* Wars cover 114, 171 and 131 of the 200 periods in the three rbar 0.16 seeds; between wars a firm moves toward its own target-return price within 2.5 % of the leader. Posted prices are

nearly equal for the two labels (t = 1–25: 84.1 against 84.6 in seed 3), but the anchors part: the unit-cost firm's own uc + 0.16 κ rises (100.5, 106.5, 122.2 over t = 1–25, 26–75, 76–125) while the profit-rate firm's falls (99.3, 96.3, 95.8), so in every pause the unit-cost firm sits at the top of the band.

- *Which technique each rule orders.* From t = 1–5 the profit-rate firms order D1 and the designs drawn from it (seed 3: D1 at t = 1, T2, D1, T7); the unit-cost firms order every design (T3, T5, T6, T9 ...), including the conflict-zone ones, whose capital per unit climbs from 160 to 359. The two labels reach almost the same unit cost (74.8 against 75.5, then 64.7 against 63.2): the extra capital buys the unit-cost firm almost no cost edge.
- *The unit-cost firm's margin.* Its margin per unit is about the profit-rate firm's (9.2 against 9.1 over t = 1–25), but over a capital twice as large late in the run its rate falls (7.3 % against 9.1 % over t = 76–125 in seed 3; −0.67 margin and 5.9 % in seed 5, where the price falls below its average unit cost). With capital per unit rising faster than its profit, each unit of retained profit buys it less capacity than its rival's; obsolescence then retires its old plants in a war (126–128 in seed 3, 131–133 in seed 4, 77–79 in seed 5: 12, 9 and 6 plants retired with the price below their unit cost for two periods).
- *Why entry stops.* The realised rate on plants built in the last five periods averages 5.3 %, 7.2 % and 6.9 %, against rbar = 16 %: the entry probability is zero in 200, 199 and 197 of the 200 periods. At rbar 0.05 (seed 3, figure B1.6) the same rate (4.2 %) is above rbar often enough for entry in 75 periods, 59 of 61 entrants inherit the unit-cost rule, and the profit-rate label has no capacity left after t ≈ 26.

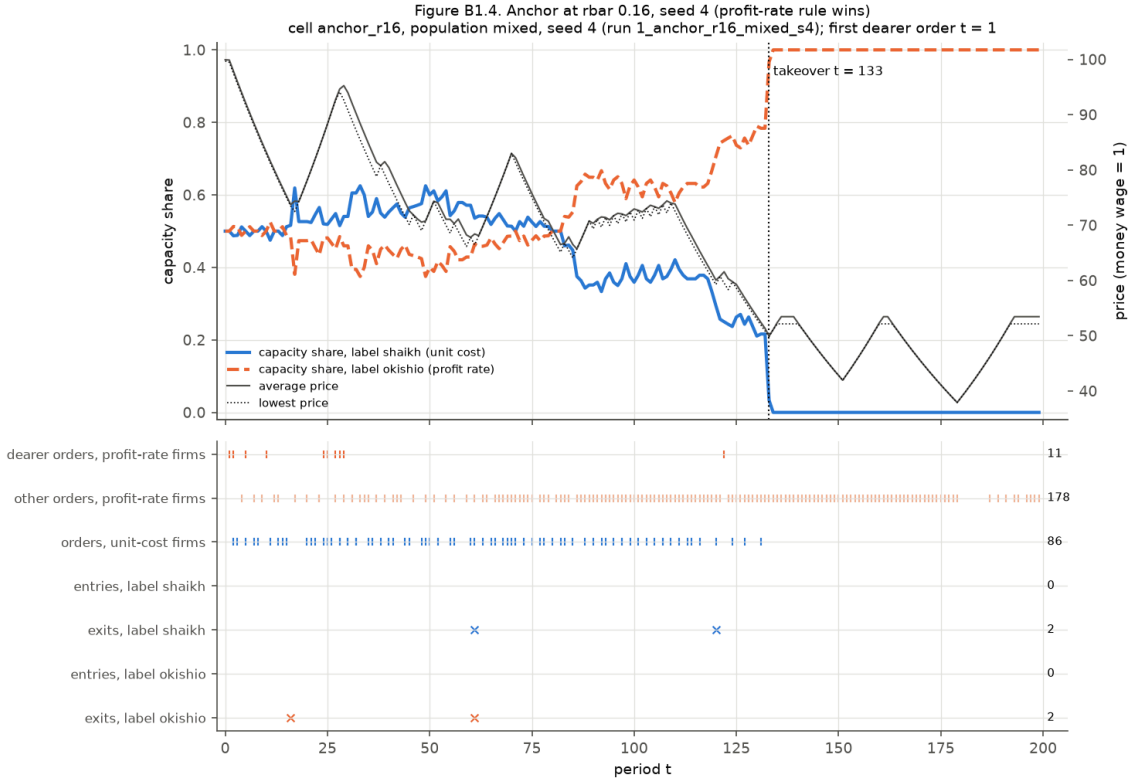


Figure B1.4 — Anchor reversal, seed 4 of `anchor_r16`.

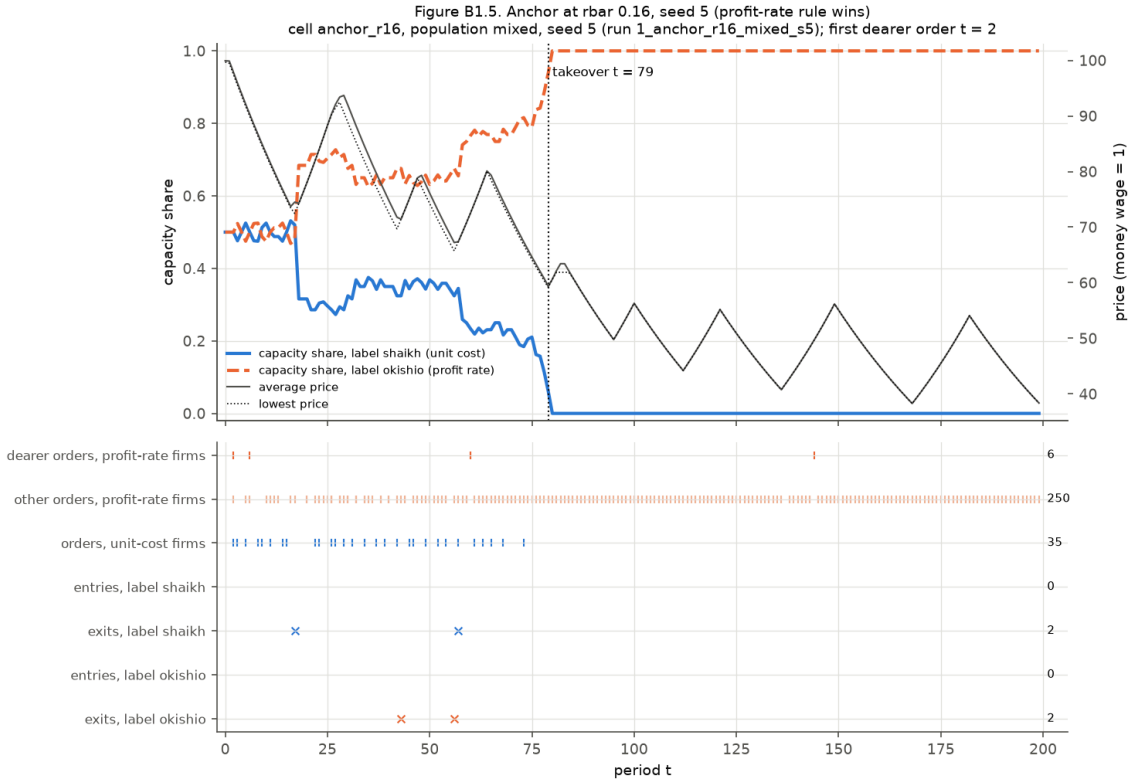


Figure B1.5 — Anchor reversal, seed 5 of `anchor_r16` (the price falls below the unit-cost firm's average unit cost).

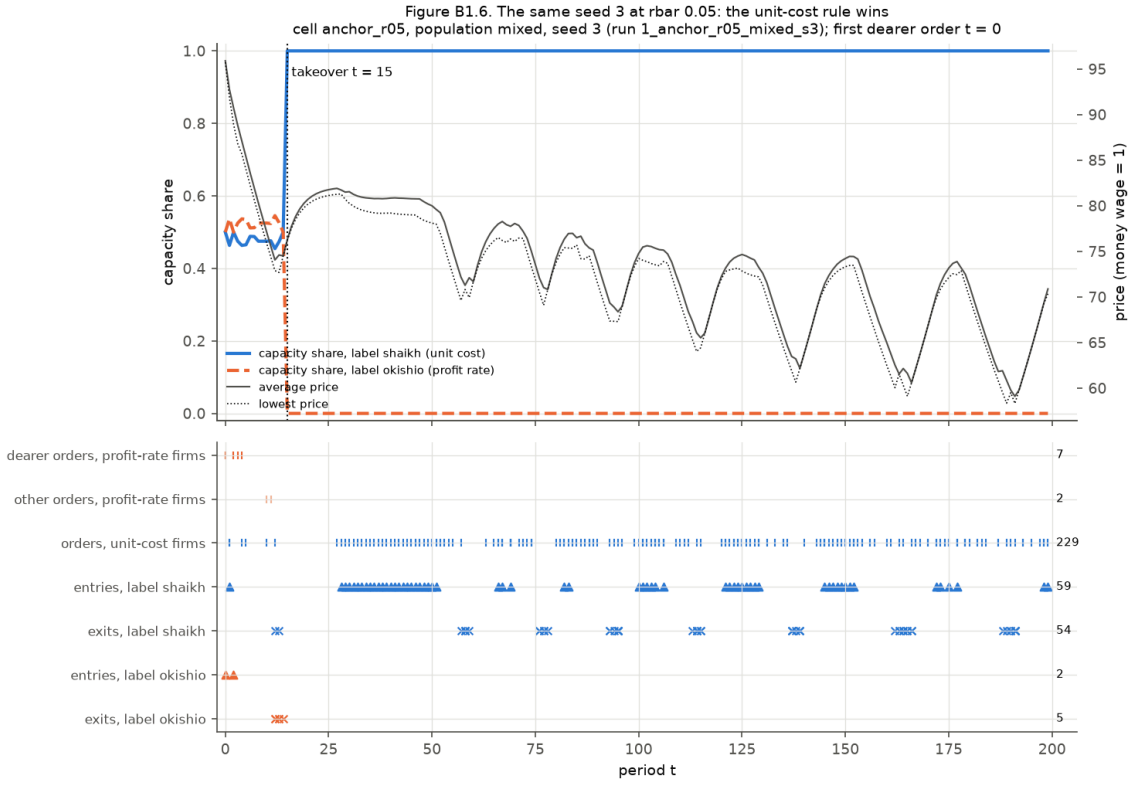


Figure B1.6 — The same rule at rbar 0.05 (seed 3 of `anchor_r05`): entry stays open and the unit-cost rule wins.

The finance twin (figure B1.7, seed 1 under retained earnings and under the accelerator with borrowing). Under retained earnings the two labels spend alike until t ≈ 100, then the profit-rate label's spending grows without bound (it reinvests every profit into a growing market) while the unit-cost label's stops at t ≈ 145; the price falls every period (98 → 48). Under the accelerator the profit-rate label spends almost nothing (10 orders in 200 periods: it loses sales, so it needs no capacity), the unit-cost label spends in bursts after shortages, and the price climbs back to

103–118. **Debt is zero in both** (at most 6e-11): not one of the 500 seeds of `lf-acd-g01-cat-r16` has a debt-to-sales ratio above 1e-6, against 0.83–0.95 of seeds under real competition with borrowing, where about 1.9 firms go bankrupt per run. What changes for the profit-rate firms is therefore not borrowing but the volume rule: under the accelerator they stop investing when they stop selling. The B2 cell without borrowing confirms it: `lf-acc-g01-cat-r16` [4, mixed, 1–200] 0.84 (0.79–0.89), controls 0.54 / 0.50, the same as the twin with borrowing.

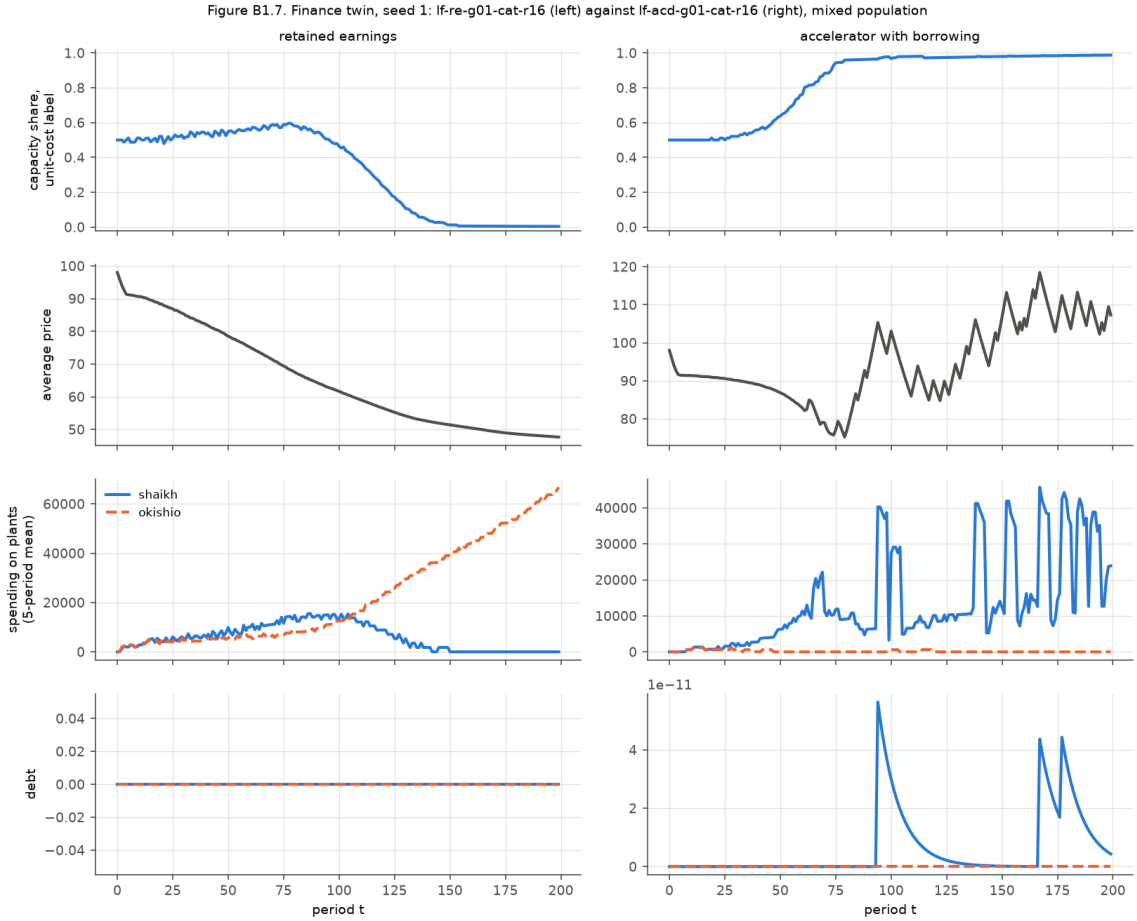


Figure B1.7 — The finance twin side by side (seed 1): retained earnings (`lf-re-g01-cat-r16`) and the accelerator with borrowing (`lf-acd-g01-cat-r16`).

Takeover timing (figure B1.3, `b1_takeover.csv`). Share of seeds with a takeover, share of takeovers before t = 100, median `takeover_t`: `base` [3, 1–500] 0.88, 0.44, 107; `lf` [3, 1–500] 0.92, 0.30, 114; `is_marx_15` [1, 1–200] 0.53, 0.18, 139; `anchor_r16` 0.94, 0.21, 127; `lf-re-g01-cat-r16` [3, 1–500] 0.99, **0.75**, 82; `lf-acd-g01-cat-r16` 0.84, **0.85**, 78; `rc-acd-g01-cat-r16` [2, 1–100] 0.25, 0.80, 70. In the cells that decide the finance question most takeovers are over before the window opens, so the window statistics (prices, profit rates, `d_share_*` without `_all`) describe the industry *after* selection, not the selection itself. Use the `_all` terms for attribution, and read window profit rates as the winners' and losers' rates in the industry the takeover left behind.

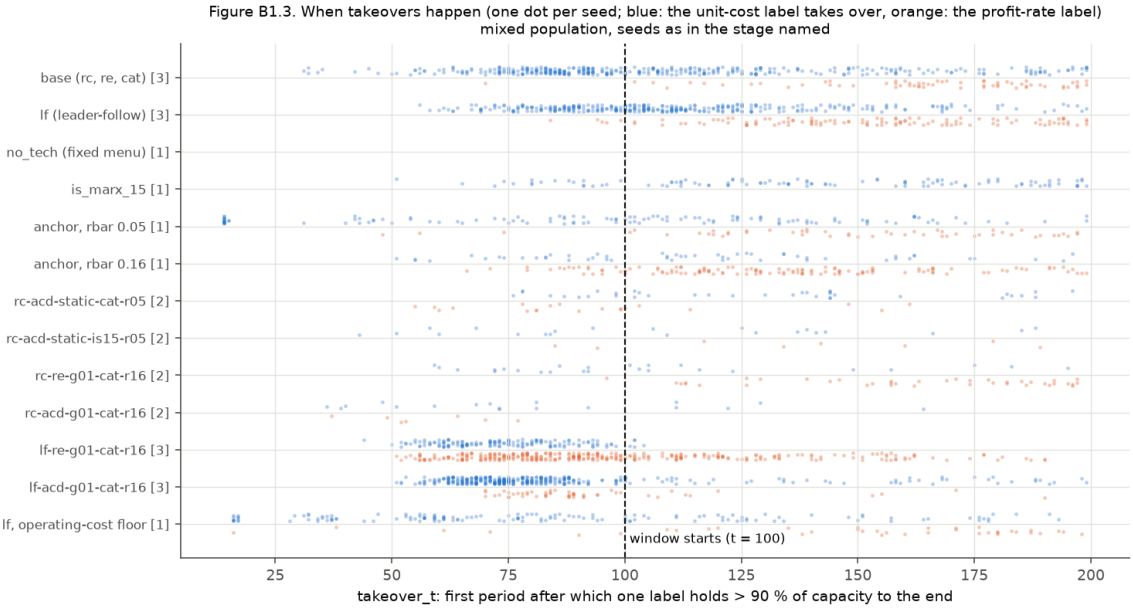


Figure B1.3 — Takeover timing in the cells of section 6.

B2. More cells

Cell [4, mixed, seeds]	Share won (95 %)	Controls (uc / pr)	Beyond drift	Reading
`rc-re-g01-cat-r16` (1–500)	0.43 (0.39–0.47)	0.49 / 0.49	**no**	the real-competition twin is *not* a reversal: 0.32 at 100 seeds was a draw from the tail
`anchor_r08` (1–200)	0.55 (0.47–0.62)	0.53 / 0.51	no	
`anchor_r10` (1–500)	**0.31** (0.27–0.35)	0.49 / 0.53	yes, profit-rate rule	0.33 at 200 seeds
`anchor_r12` (1–500)	0.46 (0.41–0.51)	0.50 / 0.48	no	the dip is confirmed at 500 seeds
`anchor_r14`	**0.30** (0.24–0.36)	0.50 / 0.52	yes, profit-rate rule	
`anchor_r16-leader` (one anchor for all)	**0.32** (0.26–0.39)	0.49 / 0.54	yes, profit-rate rule	the reversal does not need each firm's own anchor
`anchor_r16-markup` (markup anchor)	**0.27** (0.21–0.34), 17 undecided	0.52 / 0.52	yes, profit-rate rule	an anchor that favours the lowest unit cost by construction does not undo it
`anchor_r16-acc` (accelerator)	0.47 (0.40–0.54)	0.54 / 0.55	no	the accelerator removes the reversal
`anchor_r16-minf1` (`MIN_FIRMS` = 1)	0.25 (0.20–0.32)	0.55 / 0.51	yes, profit-rate rule	the floor of two firms is not the cause
`lf-re-g01-cat-r16-flooruc` (unit-cost floor)	**0.045** (0.02–0.08)	0.50 / 0.54	yes, profit-rate rule	the strongest profit-rate result of all
`lf-re-g01-cat-r16-minf1`	0.31 (0.25–0.38)	0.54 / 0.47	yes, profit-rate rule	as the default (0.32)
`anchor_r16-ps10` (plants of 10)	**0.08** (0.05–0.13), 8 undecided	0.47 / 0.50	yes, profit-rate rule	finer plants strengthen the reversal
`lf-re-g01-cat-r16-ps10` (plants of 10)	**0.13** (0.09–0.18)	0.55 / 0.52	yes, profit-rate rule	idem
`lf-acc-g01-cat-r16` (accelerator, no borrowing)	**0.84** (0.79–0.89)	0.54 / 0.50	yes, unit-cost rule	the twin's flip is the accelerator's
`scrap_ucs` (defaults, scrapping on operating cost; 1–500)	0.55 (0.50–0.59)	0.44 / 0.49	**no** (disjoint from one control only)	the base advantage needs the version 2 scrapping test
`lf-scrapucs` (leader-follow, scrapping on operating cost)	0.66 (0.59–0.72)	0.47 / 0.44	yes, unit-cost rule	no move (`lf` 0.66 [1, 1–200]): the markup floor keeps prices above unit cost

Cell [4, mixed, seeds]	Share won (95 %)	Controls (uc / pr)	Beyond drift	Reading
`lf-flooruc-scrapucs` (leader-follow, unit-cost floor, scrapping on operating cost)	0.70 (0.63–0.76)	0.48 / 0.50	yes, unit-cost rule	down from 0.83 (`lf_floor_uc` [1, 1–200])
`is15-scrapucs` (marx 1.5, scrapping on operating cost)	0.75 (0.68–0.80)	0.47 / 0.50	yes, unit-cost rule	down from 0.94, still beyond drift
`anchor_r16-scrapucs`	**0.90** (0.84–0.93)	0.45 / 0.51	yes, **unit-cost rule**	the anchor reversal is a scrapping effect
`lf-re-g01-cat-r16-scrapucs`	**0.31** (0.25–0.38)	0.54 / 0.47	yes, profit-rate rule	the leader-follow reversal is not
`lf-re-g01-cat-r16-flooruc-scrapucs`	0.43 (0.36–0.50)	0.51 / 0.50	no	the unit-cost floor's amplification is
`rc-acd-chi025` / `rc-acd-chi05` (static, rbar 0.05)	0.65 (0.58–0.71) / 0.65 (0.58–0.71)	0.43 / 0.46; 0.41 / 0.50	yes / yes, unit-cost rule	CHI 1: 0.64 [2, 1–100]
`lf-acd-chi025` / `lf-acd-chi05`	0.73 (0.66–0.79) / 0.84 (0.78–0.88)	0.44 / 0.44; 0.45 / 0.47	yes / yes, unit-cost rule	
`tau_005` / `tau_010` / `tau_015`	0.77 (0.70–0.82) / 0.79 (0.73–0.84) / 0.81 (0.75–0.86)	0.48 / 0.47; 0.48 / 0.49; 0.48 / 0.48	yes, unit-cost rule	no move (`base` 0.81, `tau_025` 0.78)
`is15-tau_010` / `is15-tau_025`	0.98 (0.95–0.99) / 0.96 (0.92–0.98)	0.53 / 0.48; 0.53 / 0.47	yes, unit-cost rule	no move (`is_marx_15` 0.94)



Figure B2.2 — Stage 4 cells: share won by the unit-cost rule, with the two controls' intervals.

The anchor threshold (figure B2.1). With rbar 0.05 (stage 1) 0.66 and 0.26 at 0.16, the share won falls as rbar rises, but not monotonically (0.55, 0.31, 0.46, 0.30 at 0.08, 0.10, 0.12, 0.14; the cells at 0.10 and 0.12 were taken to 500 seeds and the dip at 0.12 stays), and it crosses below the controls between 0.08 and 0.10. Two things change with rbar under the anchor rule: the ranking of target-return prices (on the menu D1 undercuts D2 above rbar ≈ 0.064, C undercuts D2 above ≈ 0.147) and entry (entrants in the window: 15.4 at 0.05, 2.9 at 0.08, 1.1 at 0.10, 0.5 at 0.12, 0.1 at 0.14). The common-anchor cell (0.32) and the markup-anchor cell (0.27), where the first channel is shut, still reverse, and the accelerator cell (0.47) does not: **the reversal sets in as entry closes and needs investment out of retained profits**; it is gradual in rbar but quick between 0.05 and 0.10. The briefs' anchor trap is real on the menu but plays no visible part here.

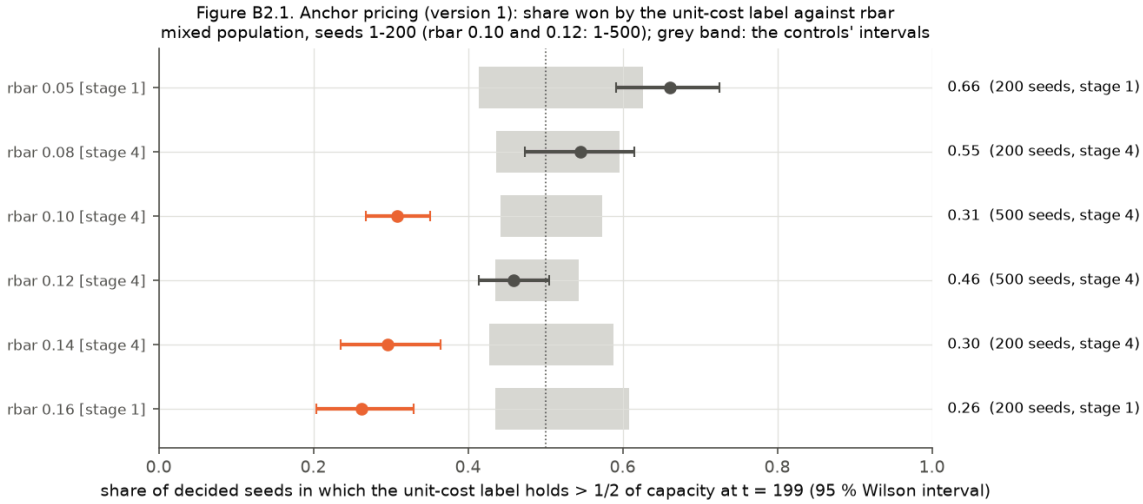


Figure B2.1 — The anchor threshold: share won by the unit-cost rule against rbar under anchor pricing, with the controls.

The scrapping test, a factor stage 1 did not vary. Version 2 retires a plant after two periods with the price below its *unit* cost ucs + δκ (`SCRAP_COST = "unit_cost"`; version 1 used the operating cost ucs, the margin of Salter, pp. 64–65, and of Shaikh's Table 7.3, row 1). Stage 1 varied only the number of periods (4 / 4: no move). With the operating-cost test the defaults lose their result: `scrap_ucs` 0.55 at 500 seeds, at drift, because old plants stay, prices fall to operating costs (the profit-rate label's hurdle averages 0.004), the industry rate falls below 5 %, entry stops (0.1 entrants in the window, 2.3 firms at the end) and the two rules choose alike. Under the innovation set (marx 1.5) the advantage shrinks (0.94 → 0.75) but survives; under leader-follow nothing moves (0.66), because its markup floor keeps prices above every plant's unit cost and the test seldom binds. The anchor reversal becomes a unit-cost win (0.90). So the base advantage of the unit-cost rule and the anchor reversal both run through the same mechanism: the unit-cost test retires plants with a large capital charge first, and in a war those are the unit-cost label's older plants when they are not the newest best practice (anchor), or the rivals' (base, where the unit-cost label holds the newest plants and the exits fall on the profit-rate firms).

What the reversals have in common, and what they do not. Both live where (1) entry is closed (rbar 0.16: the industry's rate is above rbar in fewer than 3 of 100 periods), (2) investment is financed out of retained profits, so that capacity grows by retained profit over capital per unit of the new plant, which the unit-cost rule keeps raising (both disappear with the accelerator: 0.47 and 0.85), and (3) the technical-change rule is the categories rule, under which the profit-rate rule seldom orders a dearer technique (`okishio_dearer` 0.06–0.10) while the unit-cost rule takes every conflict design, gaining little cost and much capital. They part on the scrapping test: the anchor reversal needs it (wars push the price below the unit cost of the unit-cost label's capital-heavy plants, which obsolescence then retires: 0.90 for the unit-cost rule with operating-cost scrapping), the leader-follow reversal does not (0.31 either way). Stage 2 says each condition is needed in the leader-follow corner (100 seeds): open entry (`lf-re-g01-cat-r05`) 0.64; static demand (`lf-re-static-cat-r16`) 0.51; the accelerator (`lf-acd-g01-cat-r16`) 0.83; the innovation set (`lf-re-g01-is15-r16`) **0.98**. The unit-cost floor makes it stronger (0.045), not weaker. A rerun of its seed 1 shows how: the unit-cost label holds 0.55–0.63 of capacity until t = 150, with margins and profit rates close to the profit-rate label's (6.8 against 7.0 per unit, 8.0 % against 8.6 % over t = 100–149) but plants far more capital-intensive (402 against 236 per unit); a war of ten periods then takes the price from 58 to 49 at t = 163, down to the leader's best unit cost, and 97 plants of the unit-cost label (none of the profit-rate label's) are retired by obsolescence within 15 periods, because the version 2 scrapping test compares the price with the plant's *unit* cost ucs + δκ, depreciation included: the capital-heavy vintages fail it first. Its capacity share falls to 0.02 by t = 199. Shaikh (Table 7.3 row 1, p. 271: price cutting is limited by operating costs) and Salter (pp. 64–65: old plants run until their operating cost reaches the total cost of best practice) put the margin at operating cost; the cells `*-scrapucs` of B2 test whether the reversals depend on this choice. They split the two reversals: with scrapping on operating cost the anchor reversal turns into a clear win for the unit-cost rule (0.90), the unit-cost floor's amplification disappears (0.43, at drift), but the leader-follow reversal stays (0.31). **The anchor reversal is a**

scrapping effect; the leader-follow reversal is an accumulation effect. This is Nelson and Winter's selection through profitability-keyed investment (pp. 17, 19) in the one corner where the price channel is silent; it is also the condition Shaikh's mechanism needs and does not get, a cost-cutter able to expand (p. 314).

Finance × CHI. The briefs' second version of the χ trap: with borrowing, weak selection should let the higher-rate firms out-accumulate the cheaper ones. It does not: with the accelerator and borrowing the unit-cost rule wins 0.65 at CHI 0.25 and 0.5 under real competition (0.64 at CHI 1) and 0.73 and 0.84 under leader-follow, all beyond drift. Under real competition with borrowing the profit-rate firms' hurdle is almost zero (mean 0.013 at CHI 0.25): prices sit close to costs, so the two rules nearly agree, and what remains is the unit-cost rule's edge. The accumulation channel does not need weak selection; it needs self-finance, growth and a closed industry (above).

TAU. As the paper argument of A.8 predicts, τ from 0 to 0.25 moves nothing under either technical-change rule (categories 0.77–0.81, marx 1.5 0.96–0.98): drawn designs keep a conflict zone whatever τ.

B3. Rules under which the profit-rate rule can dominate

On paper first. The research question asks *under which conditions* each rule wins, so a map of where the profit-rate rule dominates belongs to the answer, as long as the condition is a feature of the economy and not a construction that hands it the win.

1. *The price at which the hurdle is evaluated* (`choices.md`: "own"). Okishio's theorem compares techniques at prices of production (Shaikh 2016, p. 320; Shaikh 1978, pp. 241–242; Duménil and Lévy 2005, p. 209: prices of production are assumed to prevail); Shaikh insists that actual choices are made at actual and projected market prices (p. 321), and Duménil and Lévy's test evaluates next period's technology at the current wage (2005, pp. 210–211), i.e. at going prices. Shaikh's own argument says where evaluation at a lower price leads: there is a range of prices at which the lower-cost technique also has the higher rate (pp. 315–317, fig. 7.22), so any rule evaluated at a price low enough converges on the unit-cost rule. **Trap it is near**: none (it changes what the rule sees, not what it is). Implemented as `HURDLE_PRICE = "regulating"`: the profit-rate rule evaluates its hurdle at the price of production uc + $\bar{r}\,\kappa$ of the lowest-unit-cost technique in operation (the regulating capital, Shaikh 2016, pp. 265–266). The leader's price was considered and set aside: under both pricing rules followers post the leader's price in most periods, so it would hardly differ from "own".
2. *A hurdle at the general rate* (h = $\bar{r}$). It is Sraffa's criterion of choice (the lower price of production at the ruling rate, Shaikh 2016, p. 320 n. 36), Salter's replacement margin with a normal return (1966, pp. 56–57, 84–85), and the K+S payback rule (min p + b c, Dosi and Roventini 2025, pp. 37–38) read as δ + h = 1/b. It stops being "the best rate at the current price": it is the profit-rate rule evaluated at prices of production of its own best technique (at p = min_j (uc_j + $\bar{r}$ κ_j) the best rate on offer is exactly $\bar{r}$). **Trap it is near**: the anchor trap, if combined with target-return pricing (the same criterion in the price and in the choice). Implemented as `HURDLE_TARGET = "rbar"`.
3. *Weak selection with borrowing* is a condition, covered by B2's finance × CHI block.
4. *Demand where capacity is the constraint.* Stage 1: initially unmet demand 0.76 (beyond drift, for the unit-cost rule), unmet and growing 2 % 0.46, growing 3 % 0.43 (at drift, 500 seeds), never a reversal. Growth dilutes rather than reverses because at rbar 0.05 it opens entry wide (63.7 entrants in the window and 44 firms at the end under 3 % growth): entrants copy the rule of an incumbent drawn by market share, plus 2 % mutation, so heavy entry is a neutral copying process that pulls the population toward drift. Only when entry is closed does growth turn into the accumulation channel and reverse the outcome (B2 above).
5. *An evolving hurdle* (firms move h toward the realised rate of their recent plants): on paper only. Since the profit-rate firms realise less than they expect (4.7 % against a mean hurdle of 9.4 % in `base`), such a rule would lower their hurdle toward the realised rate, i.e. toward the unit-cost rule, as candidates 1 and 2 do; it would be a learning version of candidate 2 with the realised industry rate in place of $\bar{r}$. Prediction: drift at the defaults, no dominance. Not implemented.

Implementation (following the six steps of `CLAUDE.md`): `equations/regulating_price.py` (new) and `hurdle_at_general_rate` in `equations/hurdle.py`; the switches `HURDLE_PRICE` ("own" by default) and `HURDLE_TARGET` ("best" by default) in `parameters.py`; one helper in `model/scheduler.py` (`_hurdle`,

`_regulating_price`) called where the hurdle was computed before (step 4 for incumbents, step 6 for entrants), the regulating price taken once per period before any plant retires, so that no firm's order matters; tests in `tests/test_investment.py` (`TestHurdleAlternatives`: the regulating price on Shaikh's menu, ties by kappa not by order, the general-rate hurdle, the defaults unchanged, the label swap under each candidate) and both switches in `test_switches_run`. `firm.type` is read only where it was (the `HURDLE_RULES` lookup and the entrant's inheritance). **`python -m debug.compare_runs`**: `run.py` at the defaults, before and after the change, "identical: every shared column of every table within 1e-12", and the same for leader-follow with the innovation set. Since the two switches add two columns to the `runs` table, `experiments.sensitivity.rerun` now compares parameters one by one instead of `param_hash` when the stored row predates a parameter (rows of stages 1–3 still reproduce exactly: `1_base_mixed_s4`, `3_lf-re-g01-cat-r16_mixed_s7` checked).

Neutrality check. Both candidates change only the hurdle of the profit-rate rule: the unit-cost rule keeps h = 0; the regulating price is an industry price, the same for every firm, computed from all plants in operation in an order that does not depend on firms; h = r̄ is a parameter. Neither reads `firm.type` and neither hands one label an advantage by construction (the label-swap tests pass). Candidate 2 combined with the anchor rule would put the same criterion in the price and in the choice: not run.

Results (stage "b3", seeds 1–200, both controls; `outputs/analysis/b3/cells.csv`, figure B3.1):

Cell	Current rule (reference)	h = r̄ (`HURDLE_TARGET`)	regulating price (`HURDLE_PRICE`)
`base` (rc, re, cat, r̄ 0.05)	0.81 (0.77–0.84) [3, 1–500]	**0.48** (0.41–0.55), controls 0.48 / 0.45	**0.53** (0.46–0.59), 0.48 / 0.49
`lf`	0.71 (0.67–0.75) [3, 1–500]	**0.56** (0.49–0.63), 0.47 / 0.49	**0.48** (0.41–0.55), 0.47 / 0.45
marx 1.5 (`is_marx_15`)	0.94 (0.90–0.97) [1, 1–200]	0.76 (0.70–0.81), 0.53 / 0.53	0.78 (0.71–0.83), 0.53 / 0.53
r̄ 0.16 (`rbar_16`)	0.64 (0.57–0.70) [1, 1–200]	0.63 (0.56–0.69), 0.49 / 0.49	0.715 (0.65–0.77), 0.49 / 0.46

The eight candidate cells were run twice, on the Mac's VM (Linux, aarch64) before the connection dropped and in the session's container (Linux, x86_64): the shares won and both controls' shares agree cell for cell.

Mean hurdle of the profit-rate label: 0.094 (current rule, `base`), 0.050 (both candidates, `base`), 0.160 and 0.276 at r̄ 0.16; dearer orders 0.51 → 0.08–0.10 at r̄ 0.05.

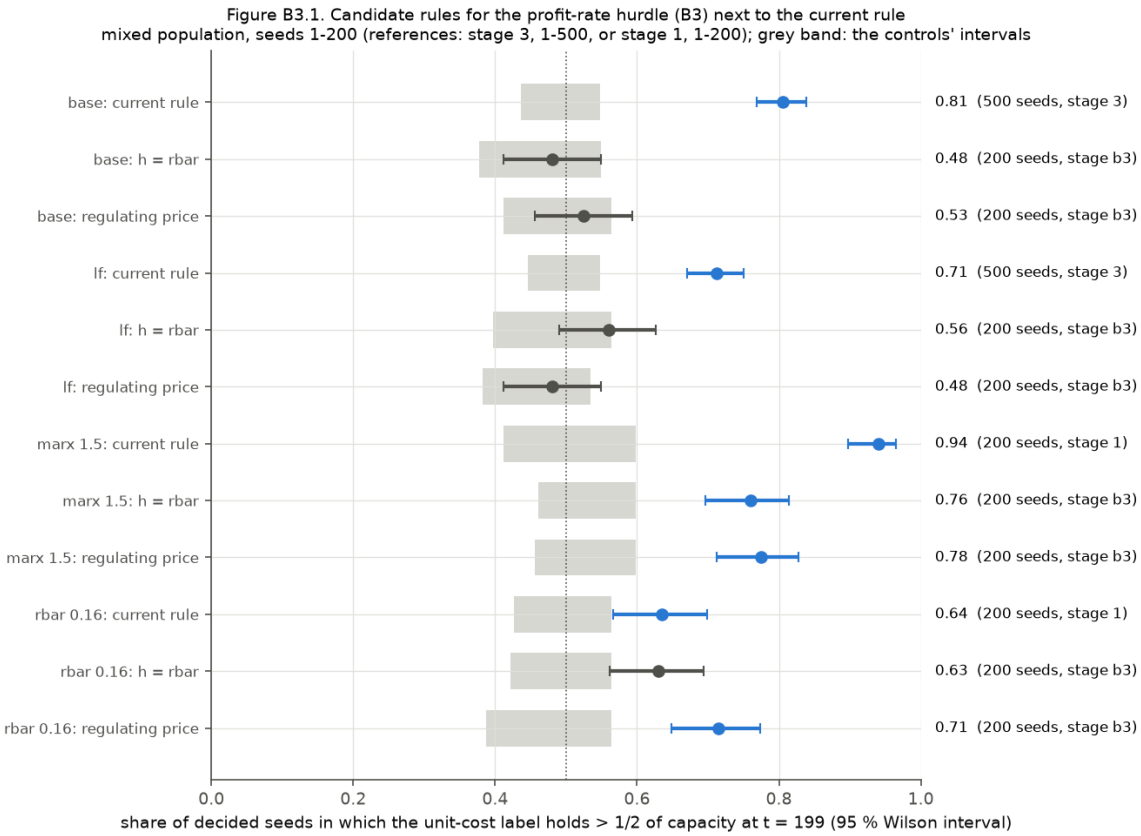


Figure B3.1 — The two candidate rules (hurdle at the general rate; hurdle evaluated at the regulating price) next to the current rule, in four cells, with both controls.

Reading. **Neither candidate lets the profit-rate rule dominate anywhere.** Evaluated at prices of production with $\bar{r} = 0.05$, the profit-rate rule's hurdle falls from 0.094 to 0.05, it stops ordering dearer techniques, and it becomes almost the unit-cost rule: the contest goes to drift under the categories rule and the unit-cost advantage shrinks under marx 1.5 (0.94 → 0.76–0.78, still beyond drift, because the Marx-biased designs keep a conflict zone at h = 0.05). At $\bar{r} = 0.16$ the regulating price is *above* the market price (prices end at 0.57 of the price of production in that cell), the hurdle rises (0.276), the profit-rate firms order more dearer techniques (0.24) and lose more (0.715). This is the clearest result of Part B for the presentation: **the unit-cost rule's advantage is the profit-rate rule's evaluation price**. Okishio's criterion applied at the prices of production of a low general rate is nearly Shaikh's criterion; applied at the inflated current price of a growing price war, or at the price of production of a high general rate the market never reaches, it chooses techniques that the falling price punishes. The industry profit rates of the controls tell the same story: under the candidates, an industry entirely on the profit-rate rule realises 8.7–9.4 % against 8.7 % on the unit-cost rule at $\bar{r}$ 0.05.

4. What this means for step 5 (proposed rows for `choices.md`, not decisions)

Proposed row	Proposed status	Why (section)
Technical change: the marx innovation set (TECH_BIAS 1.5) is the main specification; TECH_BIAS 0.5 and 3 and the neutral set are treatments; the categories rule is the version 1 robustness check; `TECH_DRAWS = False` is reported as the null (no contest)	Treatment (as the results proposed), with the reason now argued from the literature	A.2, A.3, (b)
Pricing: real competition and leader-follow are the price-war treatment; the anchor rule is not a treatment but is reported as a closed-entry, self-financed reversal (it reverses from rbar ≈ 0.10, with a common or a markup anchor too, not with the accelerator), not as the briefs' anchor trap	Treatment / reported	A.7, B2
`LEADER_FOLLOW_FLOOR`: the results recommended `unit_cost` for the leader-follow arm (0.83 against 0.66 in stage 1). Before deciding, note that in the closed growing corner the unit-cost floor *strengthens* the profit-rate rule (0.045 against 0.32)	Group decision, with this side effect recorded	B2
Finance: name the arm "investment rule" (retained earnings against the accelerator). Borrowing is inert under leader-follow (no debt in 500 seeds) and matters under real competition (bankruptcies); keep accelerator with and without borrowing as two levels if the group wants to speak of finance proper	Treatment, renamed	B1 (finance twin), B2
rbar: present 0.05 against 0.16 as *open against closed entry*, since that is what it does in the model (industry rate above rbar in 95 against fewer than 3 of 100 periods); both reversals need the closed industry	Treatment (relabelled)	A.9, B2
CHI: fixed at 1; the briefs' trap does not spring because the profit-rate firms do not realise the rate they maximise	Fixed (confirmed)	A.8
`MIN_FIRMS` = 2: no effect on either reversal	Fixed (confirmed)	B2
`SCRAP_COST`: not varied in stages 1–3, yet with the operating-cost test (version 1; Salter's and Shaikh's margin) the defaults go to drift (0.55 at 500 seeds), marx 1.5 falls to 0.75, leader-follow does not move (0.66; 0.70 with the unit-cost floor, against 0.83), and the anchor reversal flips to 0.90. Make it a factor of the next sensitivity run and argue the choice from the texts before fixing it	**Treatment to examine before step 5**	B2
`PLANT_SIZE` = 20: no move at the defaults (stage 1), but plants of 10 strengthen both reversals (0.08, 0.13)	Fixed, with the closed-industry sensitivity recorded	B2
`HURDLE_PRICE` ("own", "regulating") and `HURDLE_TARGET` ("best", "rbar")	**Candidate (analysis, 8 November 2026)**; defaults unchanged; proposed as one robustness treatment, "Okishio at prices of production", since it shows that the contest is about the evaluation price	B3
Reproducibility: Python 3.12 or later to reproduce `runs.csv` rows; CI on 3.13	Proposed	header
Window: report attribution over the whole run (`_all`), not the window, wherever takeovers come before t = 100	Proposed	B1

5. Open questions left

1. **The anchor threshold is not monotone** (0.31 at rbar 0.10, 0.46 at 0.12, 0.30 at 0.14; the first two at 500 seeds): the dip at 0.12 is not noise. A candidate explanation, not tested: on the menu the target-return ranking of C and D2 switches at rbar ≈ 0.147 and that of D1 and D2 at ≈ 0.064, so the anchors' ordering of the initial techniques changes inside the range. What differs at 0.12 is that the profit-rate firms order dearer techniques more often (0.22, against 0.12 at 0.10 and 0.14) and the accumulation term is neutral (+0.01, against −0.15 and −0.17); reruns of a few rbar 0.12 seeds against their rbar 0.10 twins would tell why.
2. **Plant size in the closed corner.** Plants of 10 make both reversals stronger (0.08 and 0.13): finer investment lets self-financed accumulation work more smoothly. The default of 20 (stage 1: 40 moved nothing at the defaults) is therefore not neutral in the closed industry; worth a sentence in the presentation.
3. **The scrapping test.** B2 found that `SCRAP_COST` decides the size of the base result (0.81 → 0.55 with the operating-cost test) and the anchor reversal (0.26 → 0.90), and it was never varied in stages 1–3. It needs its own stage 1 cell at 200 seeds under both pricing rules and the innovation set before step 5, and an argument from the texts: the operating-cost margin is Salter's and Shaikh's, the unit-cost test counts a sunk capital charge.
4. **The window.** Should the headline statistics move to t = 50–199 or to the whole run? In the cells that decide the finance question most takeovers come before t = 100.
5. **Per-mechanism random streams** (results §8.6): still open; the controls depart from 0.5 by up to 0.12.
6. **An evolving hurdle** (B3, item 5): not implemented; the prediction (drift) is cheap to test once implemented.
7. **The economy-wide Okishio question** (row 4 of section 1) cannot be answered in a one-industry model with an exogenous rbar; the controls' industry profit rates (A.6) are the nearest the model gets.
8. Documentation details: the dangling "Under the categories rule:" at the end of the 20-seed neutral-control record in `choices.md`; the 680 / 674.7 demand figure (results §8.5).

Appendix: texts opened, with the pages used

Library: Zotero (`tools/zotero.py`, libraries perso and hivemind) and the Calibre library. Pages are the printed page numbers of the PDF opened (the PDF's own page index in brackets when it differs).

Text	Library item	Pages used	For
Shaikh, A. (2016). *Capitalism: Competition, Conflict, Crises*. OUP	Zotero perso BMKVYW2U (PDF in storage 4GGIGFR6)	260–261, 265–267 (regulating capitals, figs 7.1–7.3), 271–272 (Table 7.3), 296 (turbulent equalisation of rates on new investment), 313–317 (VII.1, Tables 7.10–7.12, eq. 7.11, fig. 7.22), 318–324 (Okishio theorem, Tables 7.16–7.20, n. 36, Park 2001, figs 7.23–7.24), 328 (ch. 8 opening)	rows 1–3, 5, 6, 10; A.1–A.7, A.10, A.11, (b), B3
Shaikh, A. (1978). Political economy and capitalism: notes on Dobb's theory of crisis. *CJE* 2(2)	perso J9AEUKMD	241–242	row 4; B3
Shaikh, A. (1980). Marxian competition versus perfect competition. *CJE* 4	perso EECWZJXJ	81	row 1
Roemer, J. (1979). Continuing controversy on the falling rate of profit. *CJE* 3	perso TX2DFFI7	379–380, 382, 385, 388	rows 4–5
Duménil, G. & Lévy, D. (2005). Testing for the Marxian-classical criterion of technical choice. *Research in Political Economy* 22	perso LWGHAJ9B (the PDF attached is the whole volume)	209–213	rows 4, 6; A.5, A.7; B3
Park, C.-S. (2005). Testing Okishio's criterion of technical choice. *RPE* 22	same volume	199 (abstract only)	appendix only
Duménil, G. & Lévy, D. (1995). A stochastic model of technical change. *Metroeconomica* 46(3)	perso 48U7VRQQ	213–214, 217	rows 7; A.10, (b)
Michl, T. (2023). The falling rate of profit as a research program. *New School Economic Review*	perso FX3M6W5I	36 (abstract), 37, 40–41 (Marx-biased technical change), 46 (prisoner's dilemma, Foley 1986b), 47–48 (conflict of the two viability criteria, switchpoints R and M)	rows 2, 6; A.3, A.6, A.10, (b)
Salter, W. E. G. (1966). *Productivity and Technical Change*. CUP	perso FCUBTW7P	56–57, 64–65, 84–85 (scan, two pages per sheet)	row 9; B3
Nelson, R. & Winter, S. (1982). *An Evolutionary Theory of Economic Change*	hivemind VATF967M	17, 19, 262	row 7; A.4, A.7, B2
Foster, L., Haltiwanger, J. & Syverson, C. (2008). Reallocation, firm turnover, and efficiency. *AER* 98(1)	perso QX5VTYKC	394–395, 397	row 8; A.8
Dosi, G., Moschella, D., Pugliese, E. & Tamagni, F. (2015). Productivity, market selection, and corporate growth. *Small Business Economics*	perso TVXQ9DA7	abstract and introduction (online-first PDF, pp. 1–2; no journal pagination)	row 8
Dosi, G. & Roventini, A. (2025). *Agent-based Macroeconomics*	perso TUR32YLD	37–38 (machine choice and payback rule), 41 (entry)	B3
Silverberg, G., Dosi, G. & Orsenigo, L. (1988). Innovation, diversity and diffusion. *EJ* 98	perso EBVLD2YA	1032	row 11
Seppecher, P., Salle, I. & Lavoie, M. (2018). What drives markups? *ICC*	perso L976DQGR	HAL version, pp. 1, 6 (only the framing; not used for a claim)	appendix only

Not available or not opened, and so not cited for any claim except as reported by the texts above: Okishio (1961), *Kobe University Economic Review* (Zotero item 3ZDE32J4 has no file); Marx, *Capital* III, chs 13–15 (only vol. I is in the library; cited through Duménil and Lévy 2005, p. 209, and Shaikh 2016, p. 265); Metcalfe (1998) (not in the library); Botwinick (1993) (not in the library; cited through Shaikh 2016, pp. 265, 296); Tsoulfidis & Tsaliki (2019) (item C69II6XZ, no file; Shaikh 2016, p. 313 reports Tsoulfidis and Tsaliki 2005, 2011); Baldani & Michl (2000) (scanned PDF without a text layer, not read; cited through Michl 2023, p. 46); Foley, Michl & Tavani (2019) (the file could not be opened); Foley (1986, 1999, 2003) (not opened); Okishio (2001) and Park & Zarembka (2005) (no file); Basu (2021) and Cogliano (2022, in Calibre) are in the library and bear on the contest (Okishio's theorem; computational classical-Marxian models) but were not opened in this session. The prompt's list misses two texts the

library has and that bear directly on the contest: **Shaikh (1980)**, the reply on the choice of technique, and **Baldani & Michl (2000)**, the prisoner's-dilemma formalisation of forced adoption; and Duménil & Lévy (1995), the stochastic profitability-selection model the DLF framework comes from.